

GeoSenEsm

Application User Guide



*Geolocation and Sensor data supported
Experience Sampling Method*

Poznań, 17th August 2026

This document describes the features and capabilities of the GeoSenEsm application. GeoSenEsm enables the creation of periodic survey forms for respondents in studies that use the Experience Sampling Method (ESM). Respondents' answers are additionally correlated with their location and with sensor readings that measure conditions in their immediate surroundings or on their body. The project ships a mobile application for Android and iOS. To operate a study, researchers also need to host an administrator application for configuring the study, creating surveys and monitoring results.

GeoSenEsm — an abbreviation of *Geolocation and Sensor data supported Experience Sampling Method* — integrates geolocation and sensor data with ESM, a widely used technique that captures participants' real-time self-reports on experiences, behaviours and emotions. By pairing subjective responses with objective environmental metrics, GeoSenEsm helps urban researchers uncover more nuanced relationships between urban environments and human behaviour.

This edition of the guide covers GeoSenEsm 3.0.0, which reorganised the admin panel around a redesigned navigation menu, introduced study-wide *Survey Settings* (study logo, sending-policy calendar visibility, CSV export format), replaced the fixed Xiaomi/Kestrel sensor choice with a configurable Bluetooth Low Energy sensor-profile engine that supports any compatible device without an app update, added per-respondent sensor assignment, per-slot notification rules, an Issues overview for spotting incomplete participation, and timezone-aware scheduling for studies that span multiple time zones.

Contents

1	GeoSenEsm project	1
1.1	Project highlights	1
1.2	System architecture	1
1.3	Server deployment	2
1.4	Admin panel navigation	2
2	Survey Settings	4
2.1	General settings	4
2.2	Research area	5
2.3	Contact numbers	5
2.4	Initial survey	6
3	Choosing sensors and sensor data	7
3.1	Supported sensor types	7
3.2	Enabling sensor sources — Integrations	8
3.3	Adding a new sensor type — Sensor profiles	8
3.4	Used sensor data and sources	10
3.5	Sensor devices	13
4	Setting up survey	15
4.1	Respondents	15
4.2	Creating a survey	17
4.3	Surveys list	20
5	Monitoring and Survey results	21
5.1	Results	21
5.2	Response documents	21
5.3	Sensor data	22
5.4	Map	23
5.5	Statistics	23
5.6	Issues	28
6	Mobile application	29
6.1	Login and first-run setup	29
6.2	Home screen	30
6.3	Filling in a survey	31
6.4	Question types	33
6.5	Notifications	33
6.6	Offline mode	33
6.7	Menu and app settings	34
6.8	Additional views	38
6.9	Design consistency	42
7	Closing notes	43

1 GeoSenEsm project

GeoSenEsm is open-source software distributed under the MIT licence. The full source code lives on GitHub at <https://github.com/geosenesm>.

1.1 Project highlights

Simple, inclusive design. The main screen is dedicated to surveys only. One question is shown per page and font sizes follow the device's accessibility settings.

Data ownership and privacy. All data is stored on a server operated by the researcher. Accounts cannot be self-registered, so the researcher retains full control over who joins the study.

Scheduling and notifications. Configure start/finish times per survey and repeat them multiple times a day. Each survey defines its own set of reminder rules (e.g. "15 minutes before the slot closes"), and schedules are interpreted against each respondent's own local time zone.

Offline first. Surveys, answers, GPS points and sensor readings are stored locally when the network is unavailable and pushed to the server automatically once connectivity returns.

Admin portal. Configure the study, create surveys, manage sending policies, inspect results, and export CSV, JSON or ZIP from any modern browser.

Location and configurable sensor data. Enrich responses with GPS and with readings from any Bluetooth Low Energy sensor the researcher has configured — temperature/humidity sensors, pulse oximeters, plant-care sensors and more — without rebuilding the mobile app. Optionally constrain the study to a specific polygon.

Question types and conditional routes. Seven question types, section-level visibility rules (always, by group, conditional) and initial-survey grouping.

Respondent transparency. Participants can inspect their own submitted locations, live sensor readings and progress calendar inside the mobile app.

1.2 System architecture

GeoSenEsm is a multi-platform system based on the Flutter framework for the mobile client and on a Spring Boot / Angular pair on the server. Researchers provision their own server instance; the mobile app then connects to it using a URL supplied by the study administrator.

The server instance consists of a Java API, an admin web application, a relational database (SQL Server, storing the operational data) and a document store (MongoDB, mirroring full response documents for the download tab).

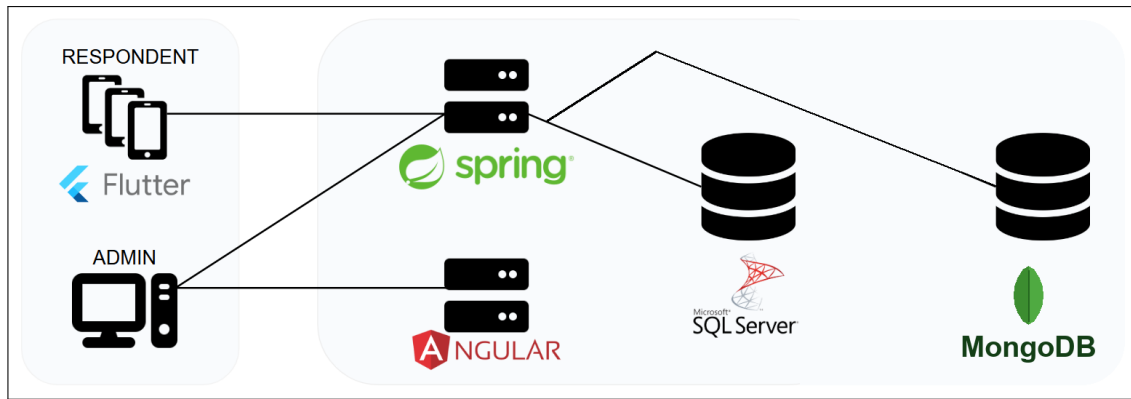


Figure 1.1: *GeoSenEsm system architecture.*

1.3 Server deployment

To set up a server, researchers need Docker installed (<https://www.docker.com/>). Docker packages applications together with their dependencies into containers; a single `docker-compose.yml` file is enough to bring up every component of GeoSenEsm without requiring specialist system-administration expertise.

Components deployed as Docker containers are:

1. the Java (Spring Boot) API,
2. the Angular admin panel,
3. SQL Server (operational database),
4. MongoDB (response-document store).

The full deployment definition and step-by-step instructions live in the DevOps repository at <https://github.com/GeoSenEsm/devops>. The only component that is not covered by `docker-compose` is the DNS / reverse-proxy configuration that maps a public domain to the API and admin containers; consult your network administrator if you need help with that step.

1.4 Admin panel navigation

The admin panel is a browser-based application hosted next to the API. Its left-hand navigation (Figure 1.2) groups every screen into six categories, which in turn map onto the next four chapters of this guide:

Settings — study-wide configuration; covered in Chapter 2, [Survey Settings](#).

Sensors (plus the Sensor Profiles and Integrations pages reached from Survey Settings) — covered in Chapter 3, [Choosing sensors and sensor data](#).

Surveys and Respondents — covered in Chapter 4, [Setting up survey](#).

Results and Statistics — covered in Chapter 5, [Monitoring and Survey results](#).

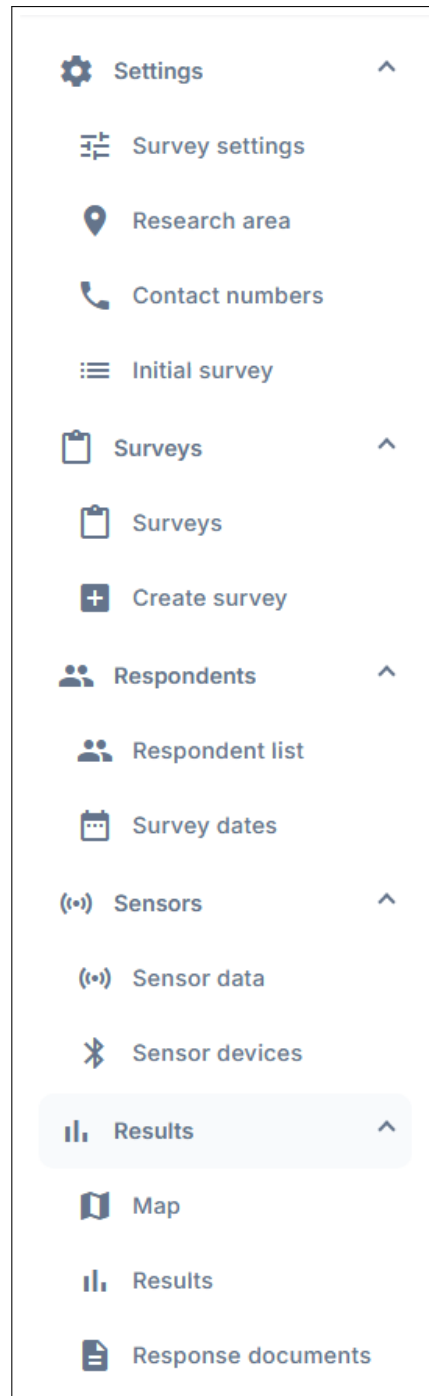


Figure 1.2: *Left-hand navigation of the admin panel, grouped into Settings, Surveys, Respondents, Sensors, Results and Statistics.*

Researchers open the admin panel in any modern browser at the URL of their deployed instance.

2 Survey Settings

This chapter covers the *Settings* navigation group: study-wide options that apply to every survey and every respondent, the research area, the contact numbers shown to respondents, and the mandatory initial survey.

2.1 General settings

The *Survey settings* page (Figure 2.1) is new in v3.0.0 and centralises options that used to have no dedicated home:

Study logo. An optional JPG or PNG (up to 1 MB) uploaded via *Upload logo* and removed with the corresponding *Remove* action. When set, it is shown at the top of the mobile app’s home screen, reinforcing the study’s identity for respondents.

Show sending policy calendar. A toggle that controls whether respondents can open a calendar of scheduled survey slots in the mobile app. Disable it for studies where the researcher would rather not reveal the exact sending schedule to participants.

CSV format. The column and decimal separators (e.g. comma vs. semicolon, dot vs. comma) used consistently for every CSV export in the admin panel and for the CSV imports used by the research-area polygon and the sensor-devices whitelist. Studies run from locales where the default comma/dot convention is unusual can switch this once, instead of reformatting every exported file by hand.

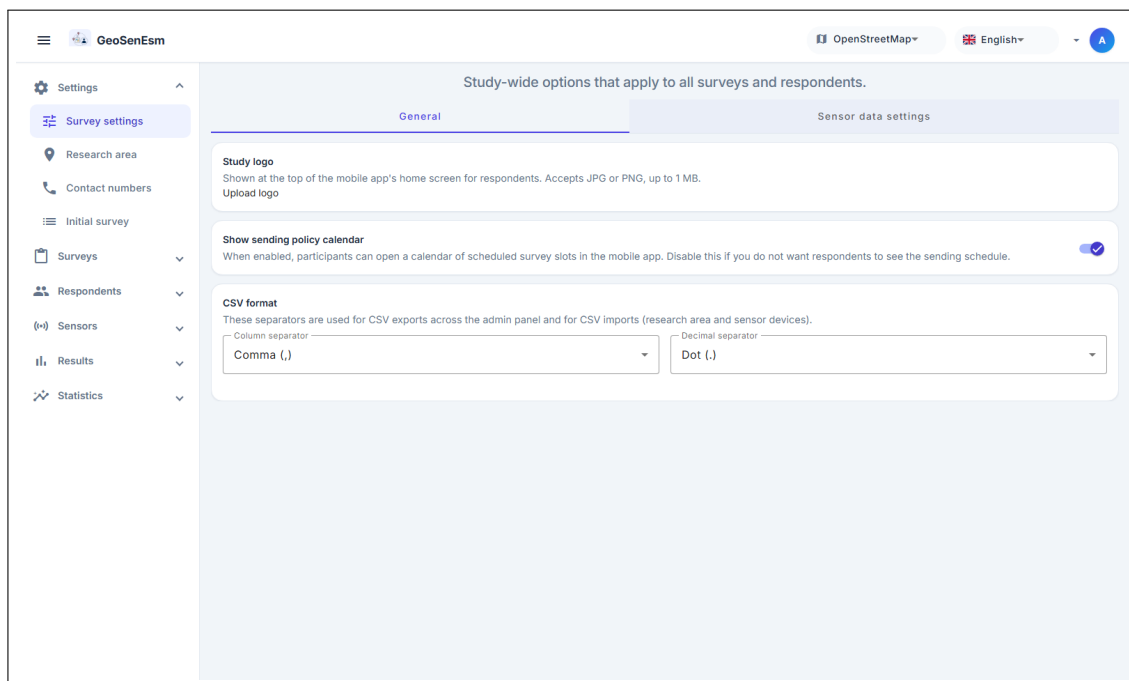


Figure 2.1: *Survey settings* — *General* tab.

The second tab on this page, *Sensor data*, configures which sensor types are collected study-wide; it is described in Section 3.4 of the next chapter, since it is tightly coupled to sensor-profile management.

2.2 Research area

Researchers can define an optional polygon representing the geographic scope of the study. Every GPS point that arrives from a respondent is compared against this polygon and marked either *inside* or *outside*. The result is used by the map view, by the *Surveys filled outside research area* chart in Statistics (Section 5.5.5) and by the `outsideResearchArea` filter on the GPS-data export.

The polygon is uploaded as a CSV file with latitude / longitude columns, formatted using the column separator configured in Section 2.1. Points must be counter-clockwise and the polygon must be closed (first and last point identical, minimum three unique points).

	A	B
1	longitude	latitude
2	16.811,52.47382065	
3	17.038,52.48762081	
4	17.015,52.35277971	
5	16.785,52.3603283	
6	16.811,52.47382065	

Figure 2.2: Example research-area CSV file.

If the research area changes mid-study, existing GPS rows are re-classified automatically.

2.3 Contact numbers

Phone numbers registered here are shown to the respondent in the mobile app so they can reach the study team (Figure 2.3).

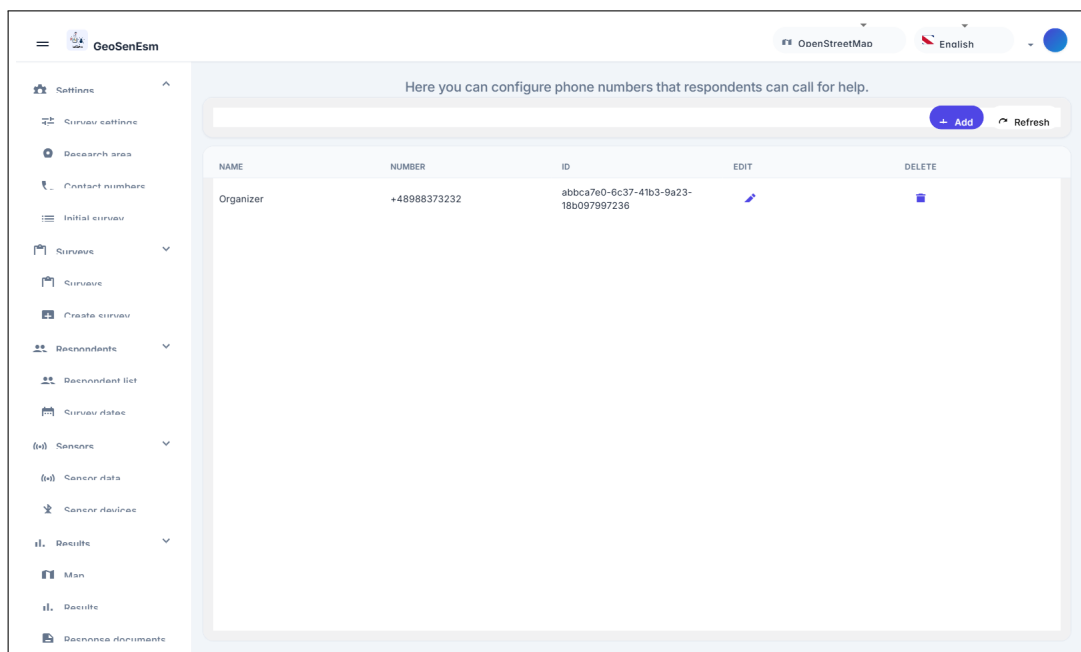


Figure 2.3: Managing study-contact phone numbers.

2.4 Initial survey

The initial survey is shown to the respondent on first login and is mandatory (Figure 2.4). Every initial-survey question must be single-choice — the selected option later becomes a column in the respondent record and can be used to route conditional or group-restricted sections in normal surveys.

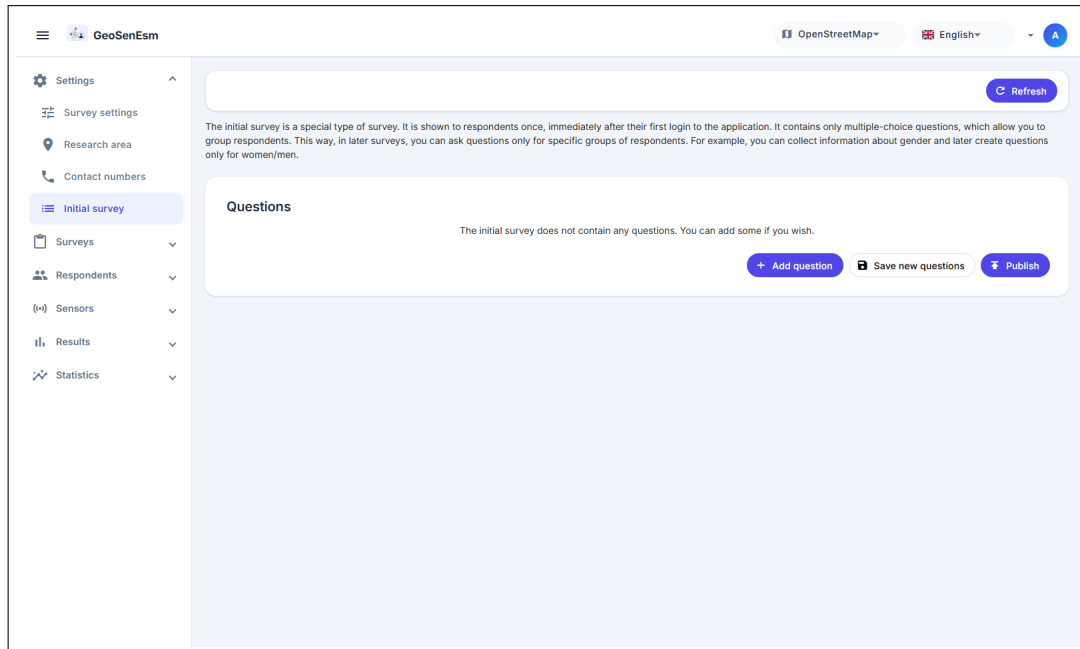


Figure 2.4: *Initial-survey editor.*

Sensor and survey setup lock

Once the initial survey is published, the study is considered live. From that point on, sensor-type management, GATT-profile edits and the sensor-data mode toggle described in Chapter 3 are locked to protect data consistency; only per-respondent sensor *assignments* stay editable, since which physical device a respondent carries can keep changing throughout the study. The same lock also engages independently of publication, as soon as the study has collected even one sensor reading — disabling a sensor type or deleting a used parameter at that point could otherwise orphan or destroy data already gathered from respondents.

3 Choosing sensors and sensor data

GeoSenEsm 3.0.0 replaces the old hard-coded choice between “Xiaomi”, “Kestrel” or “no sensor” with a configurable Bluetooth Low Energy sensor-profile engine. Researchers pick which physical sensors a study uses and which readings count as “sensor data” entirely from the admin panel; the mobile app downloads the active configuration and knows how to talk to any supported device without a new app release.

3.1 Supported sensor types

Six sensor types ship as ready-to-use templates:

Xiaomi Mi LYWSD03MMC — temperature and humidity.

Kestrel Drop D2 — temperature and humidity, with a unique serial number so no MAC whitelist is required.

Inkbird IBS-TH1 / Mini / Plus — temperature and humidity.

PC-60FW — pulse oximeter (SpO₂, pulse rate, perfusion index).

Flower Care — plant sensor (temperature, light, moisture, conductivity).

RuuviTag 4-in-1 — temperature, humidity, pressure and movement.



(a) Xiaomi Mi LYWSD03MMC.



(b) Kestrel Drop D2.

Figure 3.1: Two of the built-in sensor templates.

Every template describes its Bluetooth discovery rule, its read/notify sequence and how raw bytes decode into values as a small JSON *GATT profile*. Because the mobile app’s Bluetooth engine executes whatever profile the server publishes, a new device that speaks a similar protocol can usually be added as pure configuration — see `docs/AddingSensor.md` in the source repository for the full schema and worked examples. An app rebuild is only needed for sensors that need a transport other than Bluetooth GATT, a proprietary handshake, or a new advertisement decoder.

3.2 Enabling sensor sources — Integrations

The *Integrations* page (reached from the *Sensor data* tab of Survey Settings, Figure 3.2) is where a sensor type already installed on the server is switched on for the study. Each card shows one sensor type with an enable toggle, its connection timeout, and the used-data parameters it currently feeds. An *Available templates* section below lists any of the six built-in sensors that have not been installed yet, each with a single *Activate* button that installs the type, its raw parameter catalogue and a published GATT profile in one step.

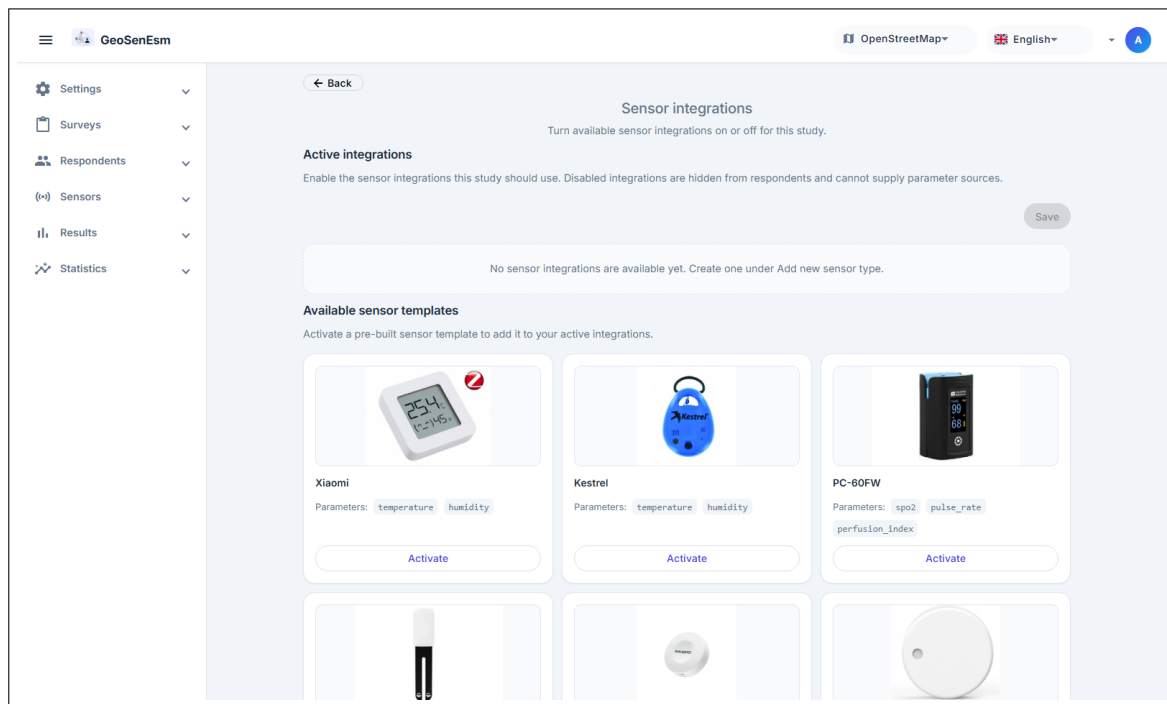


Figure 3.2: *Integrations* — enabling sensor types and activating built-in templates.

3.3 Adding a new sensor type — Sensor profiles

The *Sensor profiles* page (also reached from Survey Settings, Figure 3.3) is for sensors that are not one of the built-in templates. A new sensor type is created with a code, a display name and an integration mode (*profile*, for any device that fits the GATT engine, or *native*, for a sensor implemented as code inside the mobile app). Profile-mode types get a JSON editor where the discovery rule, read sequence and byte decoders are authored, validated against sample “golden packets”, saved as a draft and finally published. Every edit is versioned: the revision history lists past drafts and published revisions, with a rollback action for archived ones.

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Settings Surveys Respondents Sensors Results Statistics

← Back

Add new sensor type

Create a sensor type, manage bounded GATT sequences and BLE advertisement profiles, and identify types that require a compiled mobile adapter.

This is experimental functionality. Please contact the project developers if you run into any issues with it.

New sensor type

Sensor type code Sensor type name Integration mode GATT profile engine

Create sensor type

Sensor type Inkbird IBS-TH1 (inkbird_lbs_th1) — GATT profile engine Delete sensor type

GATT profile specification
The JSON spec must match the GATT profile schema. Use Validate before publishing.

Import JSON Format

Minimum engine version 1.0.0

Profile JSON

```
{
  "discovery": {
    "nameExact": "sps",
    "serviceUuid": "0000ffff-0000-1000-8000-00805f9b34fb"
  },
  "goldenPackets": [
    {
      "characteristicUuid": "0000ffff-0000-1000-8000-00805f9b34fb",
      "expected": {
        "humidity": 60.81,
        "temperature": 19.69
      },
      "packetHex": "B107C117000762"
    }
  ],
  "operations": [
    {
      "acquisition": {
        "maxPackets": 1,
        "mode": "read",

```

Figure 3.3: *Sensor profiles — a custom sensor type with its GATT profile authored, validated against a golden packet.*

Test with a real device. Below the JSON editor, a *Test with a real device* panel (Figure 3.4) lets the administrator run the draft profile against actual hardware before it ever reaches a respondent’s phone. *Connect device...* opens the browser’s own Bluetooth device picker (Web Bluetooth); once paired, the admin panel executes the draft’s write/delay/read steps exactly as written and shows each step’s raw bytes alongside the decoded value, or the specific decoder error if something doesn’t match. This only works for GATT-sequence profiles — a browser can connect to and read a named device, but unlike the mobile app’s native BLE stack it cannot passively scan raw advertisement broadcasts, so `ble_advertisement`/MiBeacon profiles rely on golden-packet validation alone. Live testing also requires a browser with Web Bluetooth support (Chrome/Edge; not Firefox or Safari).

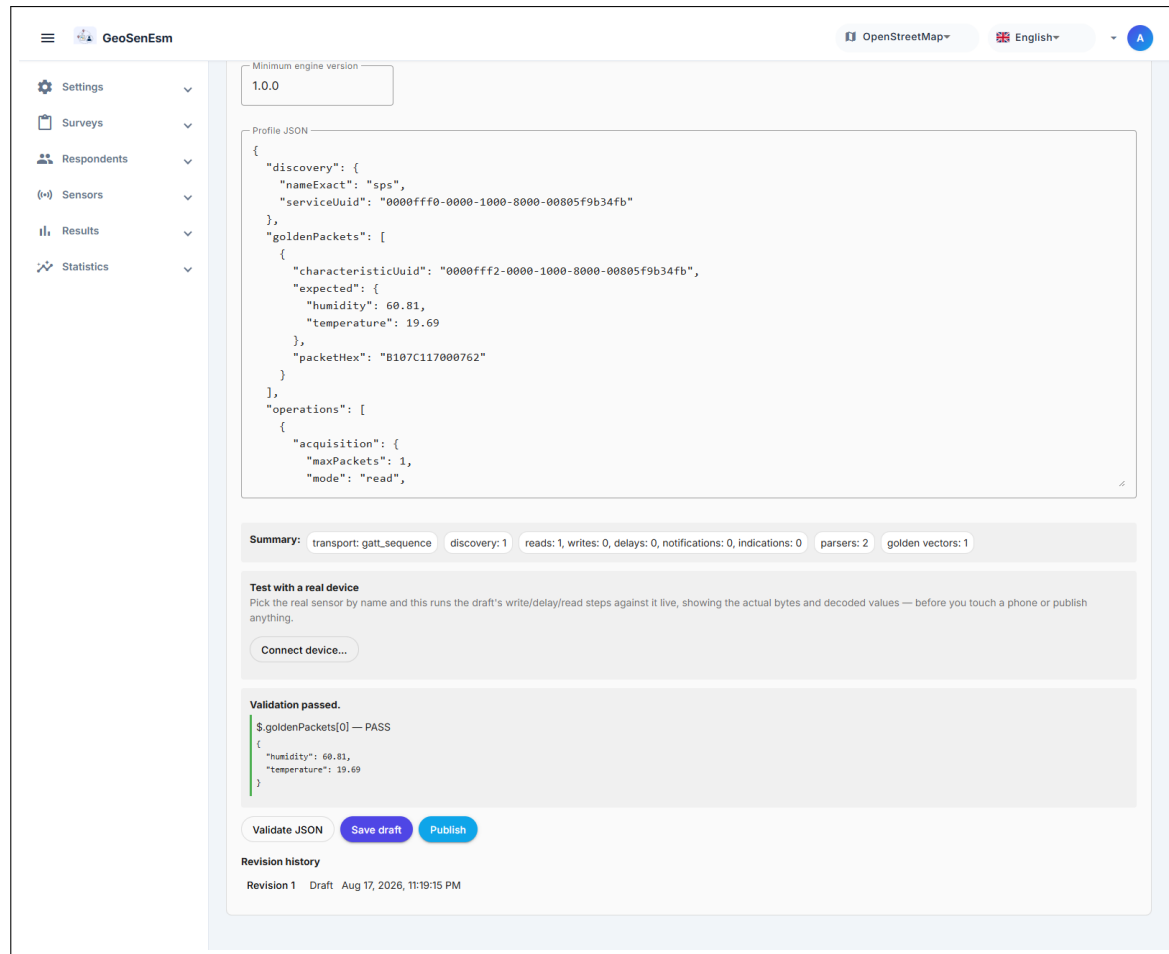


Figure 3.4: Validation summary and the “Test with a real device” panel.

3.4 Used sensor data and sources

The *Sensor data* tab of Survey Settings (Figure 3.5) decides what actually counts as “sensor data” for the study, independently of which raw sensor types exist:

Sensor data mode. A checkbox that switches the whole study between “no sensor data” and “configured sensors”.

Used parameters. Each row is one column that will appear in exports and response documents (name, data type, unit), created either from scratch or promoted directly from an enabled integration’s raw reading. A parameter can be fed by more than one raw sensor reading — for example, temperature from both a Kestrel and a Xiaomi device — each one added as an independent additional *source*. There is no ordering or priority between sources: the mobile app connects to every sensor a respondent has, and every source that actually reports a value for the survey is kept, not just the first one. Every used parameter is also guaranteed a **manual** source the moment it is created, so a respondent can always be prompted to enter a value by hand if no connected sensor covered it; unlike other sources, this one cannot be removed from the admin panel. Removing a parameter is a hard delete — there is no inactive/hidden state — and is rejected if readings have already been collected for it. Which physical device a respondent carries is assigned separately, on the *Sensor devices* tab (Section 3.5).

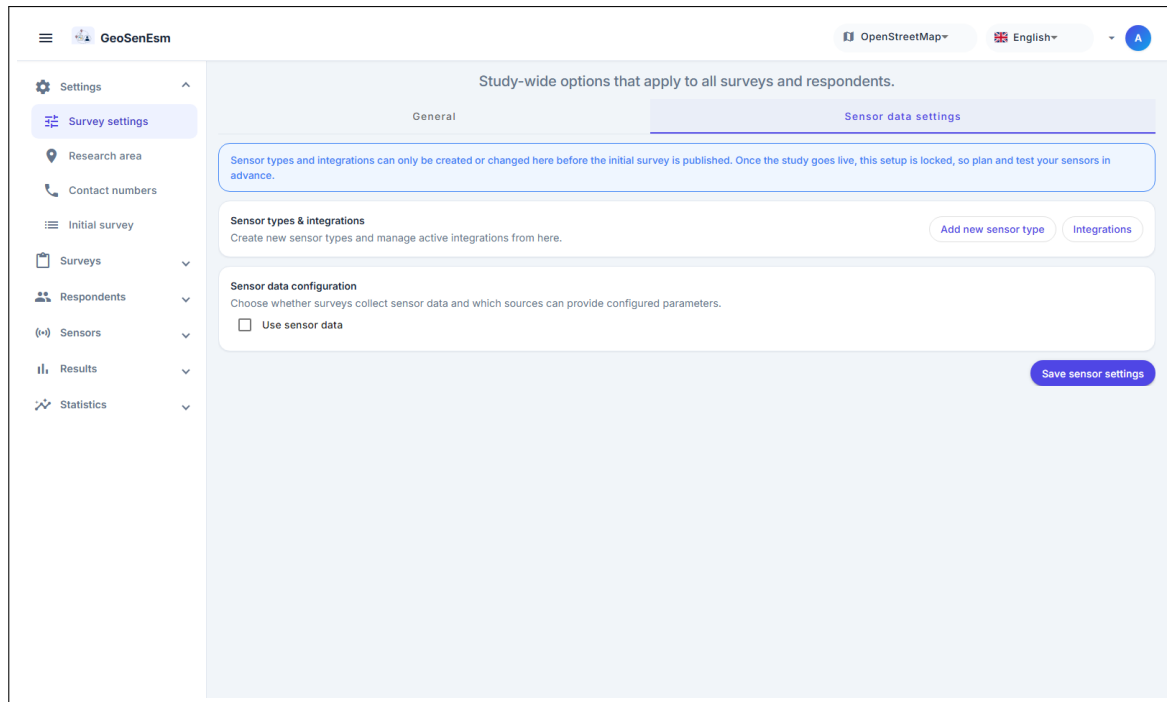


Figure 3.5: Survey settings — Sensor data tab.

Adding a new parameter. Once *Use sensor data* is checked, an *Add new parameter* panel offers the two ways to create a used parameter (Figure 3.6):

From sensor parameters. A dropdown lists every raw parameter reported by a currently *enabled* integration that isn't already in use — i.e., a raw code with no matching (name, unit) used parameter yet. Picking one and clicking *Add parameter* promotes it directly, pre-filling the new row's name/unit from the raw reading. If the raw reading's (name, unit) already matches an existing used parameter, it never appears here — instead it shows up as an option under that parameter's own *Add a new source* control (visible per-parameter once at least one source already exists), since it is a second source for the same column rather than a new column.

User-defined parameter. A blank code/name/data type/unit form for a parameter with no sensor behind it at all (for example, a manually-reported symptom score). It gets the same guaranteed `manual` source as any other used parameter and behaves identically everywhere else in the admin panel.

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OpenStreetMap English

Sensor types and integrations can only be created or changed here before the initial survey is published. Once the study goes live, this setup is locked, so plan and test your sensors in advance.

Sensor types & integrations
Create new sensor types and manage active integrations from here. [Add new sensor type](#) [Integrations](#)

Sensor data configuration
Choose whether surveys collect sensor data and which sources can provide configured parameters.
☒ Use sensor data

Sensor data columns
Define the parameter columns collected from sensors or manual entry.

CURRENT PARAMETERS

Code	Name	Data type	Unit
temperature	Temperature	Decimal	C
Save parameter Remove parameter			
Sources Manual (temperature) Xiaomi (temperature) Inkbird IBS-TH1 (temperature)			
humidity	Humidity	Decimal	%
Save parameter Remove parameter			
Sources Manual (humidity) Xiaomi (humidity)			

ADD A NEW SOURCE

Sensor type [Add source](#)

Add new parameter
Promote a raw sensor reading into a used parameter, or define one manually.

From sensor parameters
Pick a raw parameter reported by an active integration that hasn't been added as a used parameter yet.

Sensor parameter

[Add parameter](#)

User-defined parameter
Define a parameter manually, without linking it to a specific sensor reading.

Code Name Data type

Unit

[Add parameter](#)

[Save sensor settings](#)

Figure 3.6: Sensor data columns while still unlocked — an existing parameter with three sources plus one pending Add a new source, and both Add new parameter options available.

Once the initial survey is published, every mutating control on this tab locks — the sensor data mode checkbox, every used parameter's name/data type/unit fields, adding or removing a parameter, adding a source, and the sensor types/integrations shortcut buttons (Figure 3.7). A yellow banner explains why, and the parameter columns already collected remain visible (read-only) so the study's sensor-data shape stays inspectable even though it can no longer be changed.

Figure 3.7: *Sensor data columns after the initial survey has been published — setup is locked and every field is read-only.*

3.5 Sensor devices

The *Sensor devices* tab (Figure 3.8) is where a MAC address is tied to a sensor type and, optionally, to the respondent who carries it. A MAC address is itself optional: a sensor type that can be matched purely by discovery (advertised name, service, or product ID) can be assigned without pre-registering which physical unit a respondent has. A whitelist entry matters most for the Xiaomi LYWSD03MMC, whose Bluetooth name is identical on every unit: without a MAC on file the mobile app cannot tell one respondent's sensor from their neighbour's. Kestrel and other devices with a unique identifier do not need a whitelist entry to be told apart, but can still be recorded here for bookkeeping.

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Configure sensor MAC addresses and assign each sensor to a respondent so the mobile app can sync the correct device automatically.

Search + Add Refresh Export Import

ID	TYPE	MAC ADDRESS	RESPONDENT	UPDATE	DELETE
1	Flower Care		00001		
2	Ruuvi Tag 4-in-1		00001		
3	Xiaomi	A4:C1:38:0F:2B:9E	00001		

Items per page: 30 1 - 3 of 3

Figure 3.8: Sensor devices — MAC whitelist with respondent assignment.

4 Setting up survey

This chapter covers the *Surveys* and *Respondents* navigation groups: authoring a survey's structure and schedule, and managing the people who will answer it.

4.1 Respondents

The Respondents view lists all accounts, lets the administrator reset a password, and is where new accounts are generated (Figure 4.1).

USERNAME	SURVEY START	SURVEY END	TIMEZONE	ID	ACTIONS
00001			UTC	167b3ae44-1c5f-4b60-b9cc-f7d67aaf9d8e	
00002			UTC	db2024b5-7d9e-47b7-a010-c29b5a43b47c	
00003			UTC	43d2a86d-af84-4f6d-9762-7e5f0d9e5786c	
00004			UTC	8b80f9fe-dc83-43ad-9205-3af7effeb222	
00005			UTC	479eb808b-80b4-43c1-9a08-071ef13219f6	
00006			UTC	2b4fa9aa-42de-4454-a2a0-58bf2a139034	
00007			UTC	9a27b292-a0cc-4be5-a225-b2ba0269e900	
00008			UTC	0e23bf55-2d84-4586-bd13-f3ad168886e0	

Figure 4.1: *Respondents list.*

Filters include a username/ID search and a survey-window presence filter (Figure 4.2).

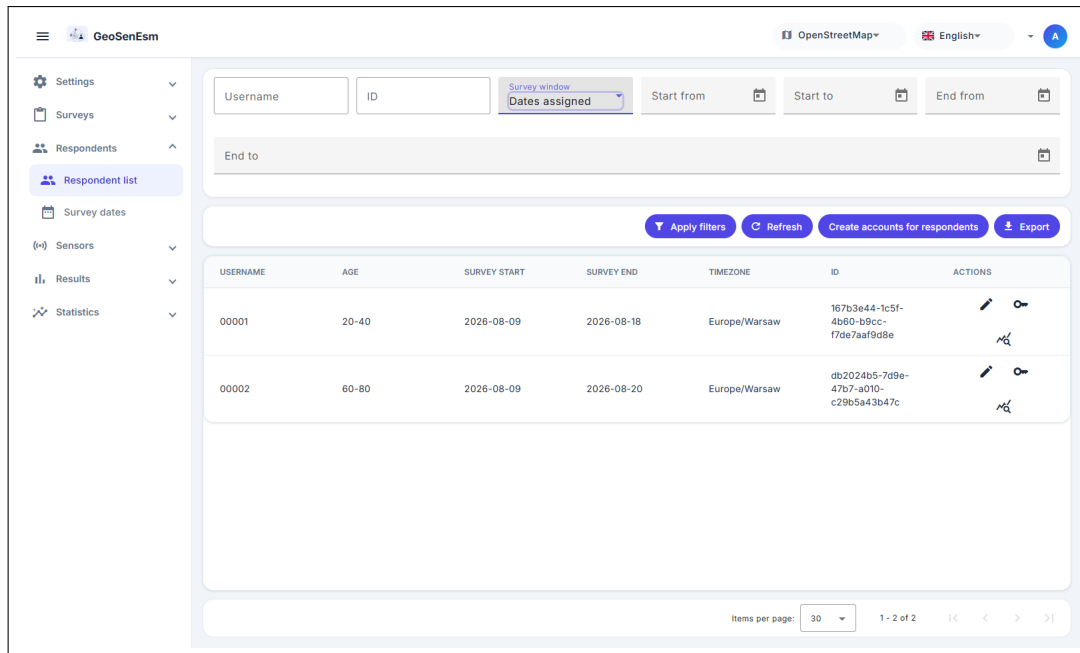


Figure 4.2: Respondent filters — here narrowed to respondents with a survey window assigned.

When a batch of accounts is created (Figure 4.3), the generated credentials are copied to the clipboard — they cannot be recovered from the server afterwards. Paste the CSV into a safe place before closing the dialog.

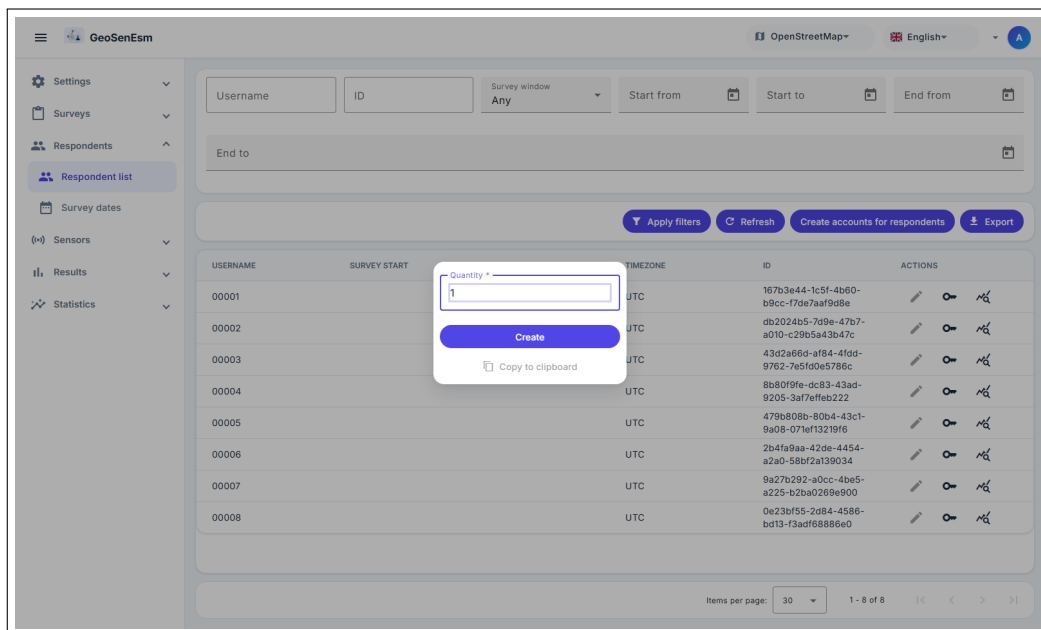


Figure 4.3: Creating respondent accounts. The generated credentials are exported via Copy to clipboard.

4.1.1 Survey dates

Each respondent can optionally be assigned a *survey window* — a start and end date bounding when they are expected to participate. The *Survey dates* page (Figure 4.4) batch-assigns or clears these dates for any selection of respondents, and shows a daily activity chart of how many respondents currently have an open window. Survey windows drive the *Active* filters used in Statistics (Section 5.5), Daily completion (Section 5.5.4) and Issues (Section 5.6).

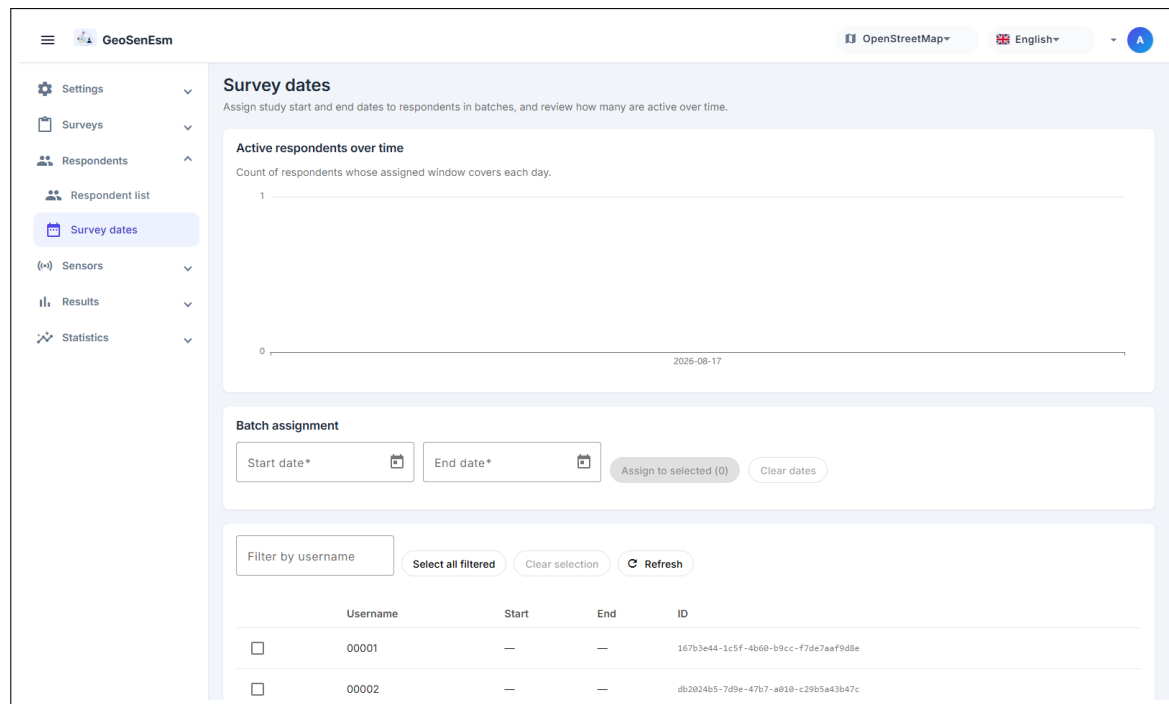


Figure 4.4: Survey dates — batch assignment and activity overview.

4.2 Creating a survey

Survey creation is broken into three steps: editor, sending policy (with notifications) and preview.

4.2.1 Editor

A survey is a title plus an ordered list of *sections*, and each section is an ordered list of *questions* (Figure 4.5). There is no separate "add question" step outside a section — every question belongs to exactly one section, and a survey needs at least one section with at least one question before it can be saved. *Add question* inserts a new question at the bottom of the section you clicked it in (or right below a specific question, via that question's own *Add question below* control); *Add section* appends a new section to the survey. Sections and questions can be reordered with their up/down arrows and removed with their delete icon — a section always keeps at least one question, so the last one in a section cannot be deleted until another is added.

One page per section or one question per screen. Each section has a *Display on one page* toggle. Checked (the default), every question in that section is presented together on a single screen in the mobile app; unchecked, the app instead pages through that section's questions one at a time, question-by-question. This only matters for sections with more than one question — a single-question section looks identical either way.

Section visibility. Each section has one of three visibility rules (Figure 4.6a):

Always visible. Shown to every respondent. The default, and the only option available to the survey's very first section.

Group visibility. Shown only to respondents whose initial-survey answers place them in a specific group.

Conditional visibility. Shown only if the respondent picked a specific option on an earlier single-choice or image-choice question.

Setting up a conditional section — what to do first. Conditional visibility is configured in two places, and the order matters:

1. **First**, on the section that should stay hidden until triggered, open its *Visibility* dropdown and choose *After choosing a specific answer*. This does not yet say *which* answer triggers it — it only registers the section as an available target.
2. **Only then**, go to the earlier single-choice or image-choice question whose answer should reveal it. Every one of that question’s options has its own *Show section* dropdown; pick the just-registered section there. Only sections positioned *after* the current one are ever offered, so a section can’t accidentally be triggered by something that comes after it.

Doing this in the opposite order leaves the option’s *Show section* dropdown empty: a section only appears as a choice once its own visibility has already been set to *After choosing a specific answer*. Different options on the same question can reveal different sections (or none), and any number of options across any number of earlier questions can point at the same conditional section.

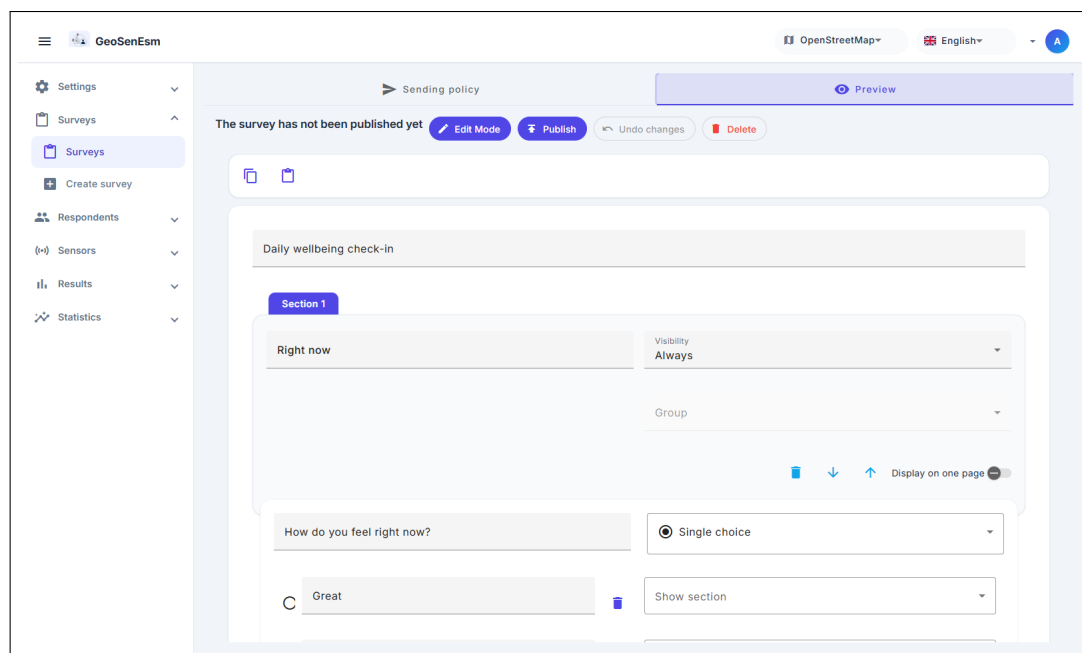
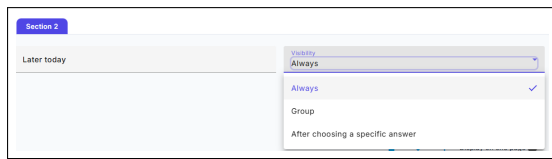
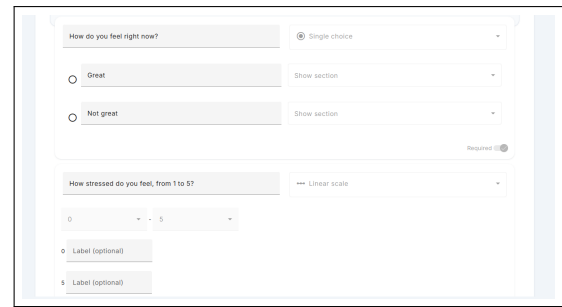


Figure 4.5: Survey editor.



(a) Section visibility rules.

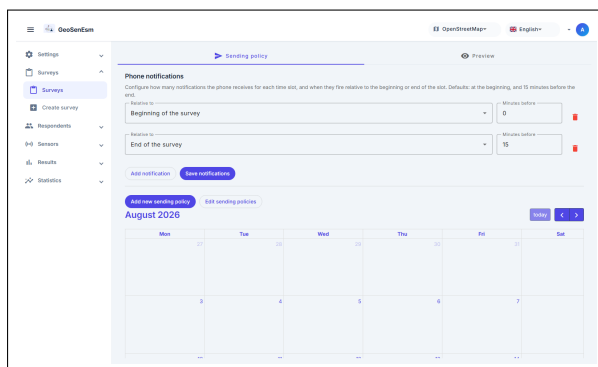


(b) Two of the seven question types (single choice, linear scale) as authored in the editor.

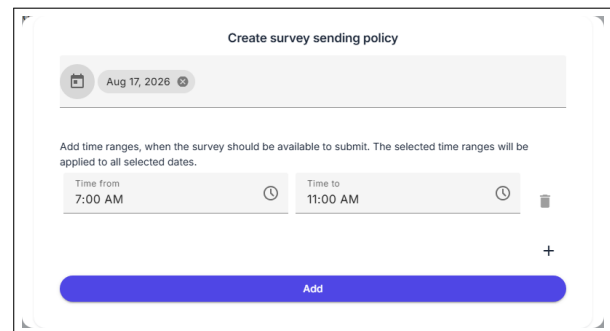
Figure 4.6: Building blocks exposed by the editor.

4.2.2 Sending policy and notifications

Once the structure is saved, the administrator defines the sending policy — when the survey will be open for participants. The schedule is drawn on a calendar view (Figure 4.7, left); the per-slot form is shown on the right. Every scheduled time is interpreted in each respondent's own local time zone (detected automatically by the mobile app at login), so a slot that opens at 9:00 reaches every participant at 9:00 local time regardless of where the study runs.



(a) Calendar overview.



(b) Per-slot form.

Figure 4.7: Sending-policy editor.

The *Phone notifications* panel (Figure 4.8) is new in v3.0.0: instead of a single fixed reminder, each survey can define up to ten notification rules, each relative to the *beginning* or the *end* of a slot and offset by a chosen number of minutes. A newly created survey starts with the previous default behaviour (a notification when the slot opens, and another 15 minutes before it closes) already filled in as editable rows.

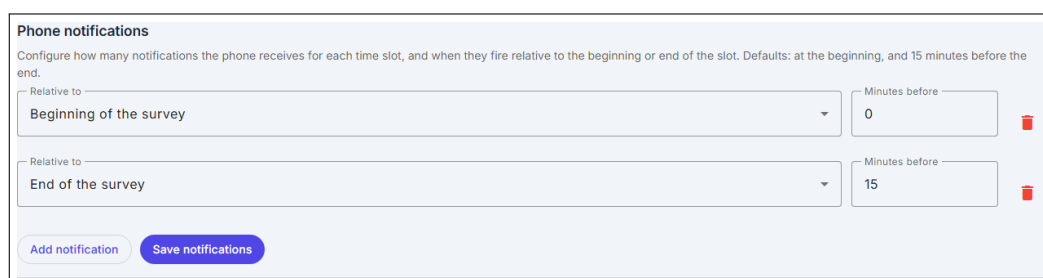


Figure 4.8: Sending policy — notification rules.

4.2.3 Preview and publish

The preview tab (Figure 4.9) is a final opportunity to spot mistakes. Once the survey is published, its structure becomes immutable — respondents can start filling it in, and further edits are no longer possible. The preview also offers a *Copy* action which is handy when reusing a complicated structure as the basis for a new survey.

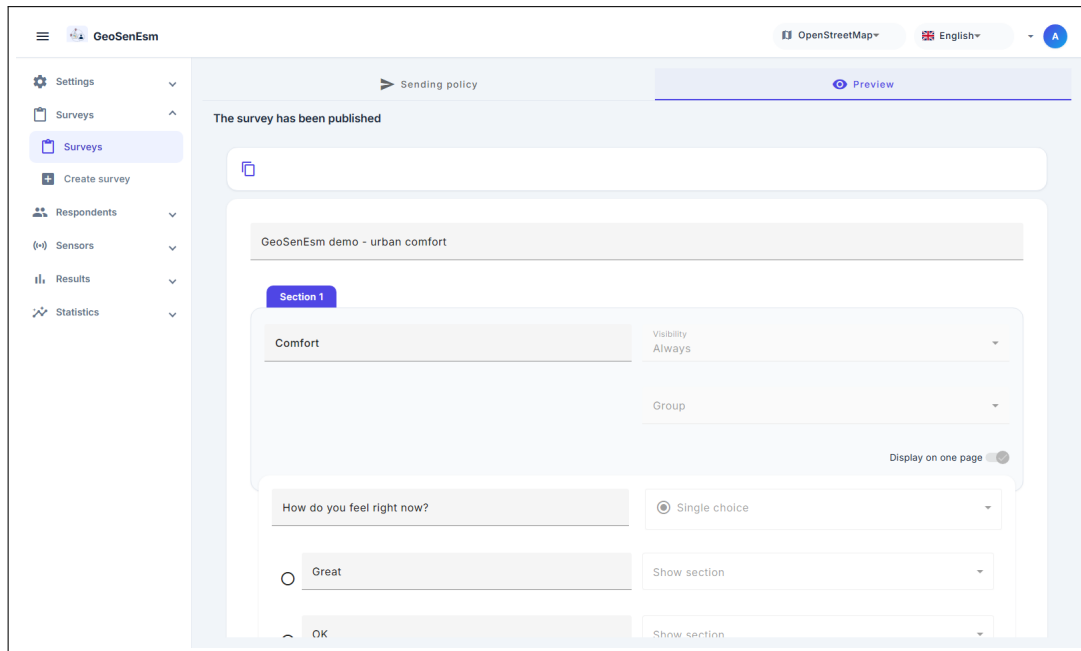


Figure 4.9: Survey preview.

4.3 Surveys list

The Surveys tab lists every survey together with its current sending policy. Schedules and notification rules can be edited from here at any time.

5 Monitoring and Survey results

This chapter covers the *Results* and *Statistics* navigation groups: everything researchers use once a study is running to inspect submitted data and spot problems early.

5.1 Results

The Results tab (Figure 5.1) shows a tabular view of all submitted answers. Rows can be filtered by survey and by respondent and exported to CSV, using the separators configured in Section 2.1.

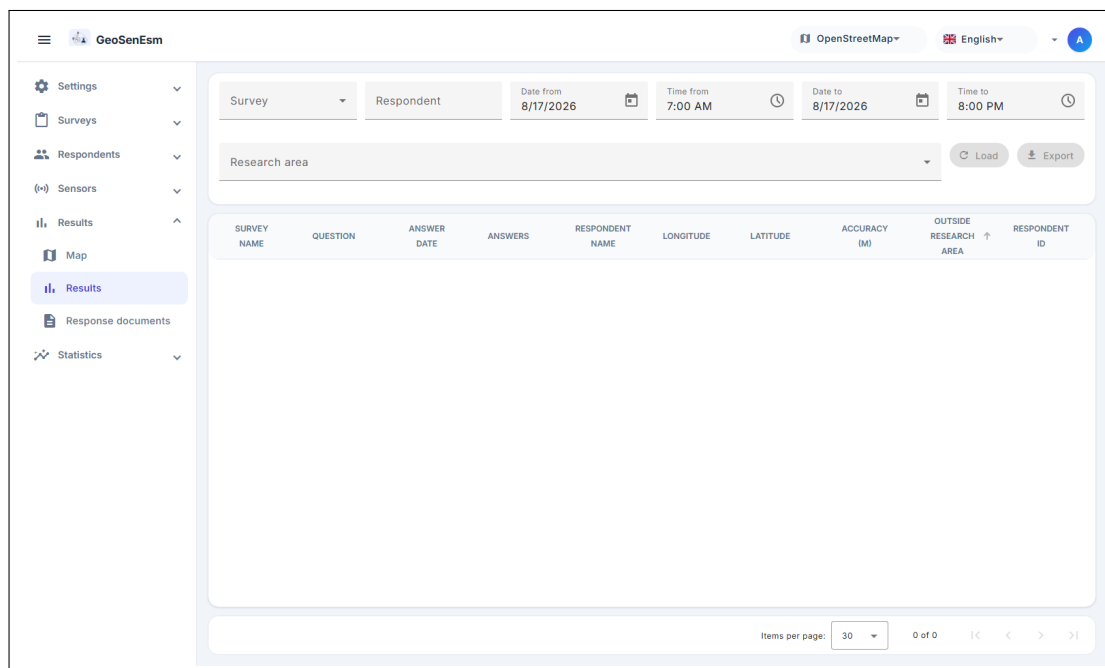


Figure 5.1: *Results tab.*

5.2 Response documents

The Response documents tab is backed by MongoDB. It contains one JSON document per submission, with the exact payload the mobile app sent to the server (answers, timestamps, one embedded sensor reading per connected sensor type, matching survey metadata). Two workflows are supported:

- **Download** one specific document from the grid by clicking the download icon in the row. The result is a self-contained JSON file that can be inspected offline.
- **Export** the full filtered set as a ZIP archive.

By default the date range is set to *today* → *tomorrow* (local time), which matches the most common use case: reviewing the responses that have just come in (Figure 5.2). The *Clear filters* button resets the range back to that default. The paginator and page-size selector stay visible even when the results table is very wide, so switching pages does not require horizontal scrolling.

PARTICIPATION DATE	SURVEY	RESPONDENT	ANSWERS	ACTIONS
8/17/26, 2:39 PM	GeoSenEsm demo - urban comfort	00003	4	Download
8/17/26, 12:55 PM	GeoSenEsm demo - urban comfort	00006	4	Download
8/17/26, 12:40 PM	GeoSenEsm demo - urban comfort	00001	4	Download
8/17/26, 11:31 AM	GeoSenEsm demo - urban comfort	00005	4	Download
8/17/26, 8:42 AM	GeoSenEsm demo - urban comfort	00002	4	Download
8/17/26, 8:06 AM	GeoSenEsm demo - urban comfort	00005	4	Download
8/17/26, 6:42 AM	GeoSenEsm demo - urban comfort	00001	4	Download
8/17/26, 2:27 AM	GeoSenEsm demo - urban comfort	00006	4	Download

Figure 5.2: Response documents tab (Mongo-backed).

5.3 Sensor data

The *Sensors data* tab (Figure 5.3) lists every sensor reading uploaded by respondents. Its columns follow whichever parameters the study has configured in Section 3.4 — temperature and humidity for a study using Xiaomi or Kestrel devices, SpO₂ and pulse rate for one using a pulse oximeter, and so on. Filters mirror the ones available for GPS data (date range, respondent).

DATE AND TIME	SOURCE	READINGS	RESPONDENT ID	SURVEY ID
8/17/26, 4:56 PM	flower_care	conductivity: 1036, light: 1472, temperature: 24.00, moisture: 54.00	167b3e44-1c5f-4b60-b9cc-f7de7aaf9d8e	-
8/17/26, 4:56 PM	xiaomi	humidity: 59.00, temperature: 23.53	167b3e44-1c5f-4b60-b9cc-f7de7aaf9d8e	-
8/17/26, 4:56 PM	ruuvi	pressure: 998.91, humidity: 56.66, temperature: 23.46, movement: 40	167b3e44-1c5f-4b60-b9cc-f7de7aaf9d8e	-
8/17/26, 4:59 PM	xiaomi	humidity: 58.00, temperature: 23.59	167b3e44-1c5f-4b60-b9cc-f7de7aaf9d8e	-
8/17/26, 4:59 PM	flower_care	conductivity: 1037, temperature: 23.90, light: 1286, moisture: 54.00	167b3e44-1c5f-4b60-b9cc-f7de7aaf9d8e	-
8/17/26, 4:59 PM	ruuvi	movement: 40, pressure: 998.93, humidity: 56.40, temperature: 23.47	167b3e44-1c5f-4b60-b9cc-f7de7aaf9d8e	-
8/17/26, 5:21 PM	flower_care	conductivity: 1039, light: 815, moisture: 53.00	167b3e44-1c5f-4b60-b9cc-f7de7aaf9d8e	-
8/17/26, 5:21 PM	ruuvi	pressure: 998.82, humidity: 55.92, movement: 40, temperature: 23.43	167b3e44-1c5f-4b60-b9cc-f7de7aaf9d8e	-
8/17/26, 5:26 PM	ruuvi	temperature: 23.43, humidity: 55.92, pressure: 998.82, movement: 40	167b3e44-1c5f-4b60-b9cc-f7de7aaf9d8e	f2fe0e6d-cb76-48bd-ad3d-8c15e4765362
8/17/26, 5:26 PM	xiaomi	temperature: 23.44, humidity: 56.00	167b3e44-1c5f-4b60-b9cc-f7de7aaf9d8e	f2fe0e6d-cb76-48bd-ad3d-8c15e4765362

Figure 5.3: Sensor data tab.

The *Survey ID* column is populated only for readings the mobile app captured while the respondent

had a survey open — those are bundled into that survey’s submission and inherit its participation ID. A reading taken any other time (a background sync, or the respondent checking a live sensor value outside of a survey) is uploaded on its own, with no participation to attach to, and shows “–” in that column instead. Both kinds of reading are the same rows in the same table; only the presence of a Survey ID tells them apart.

5.4 Map

The Map tab (Figure 5.4) plots every GPS point in the system. Points are colour-coded by whether they lie inside or outside the research-area polygon.

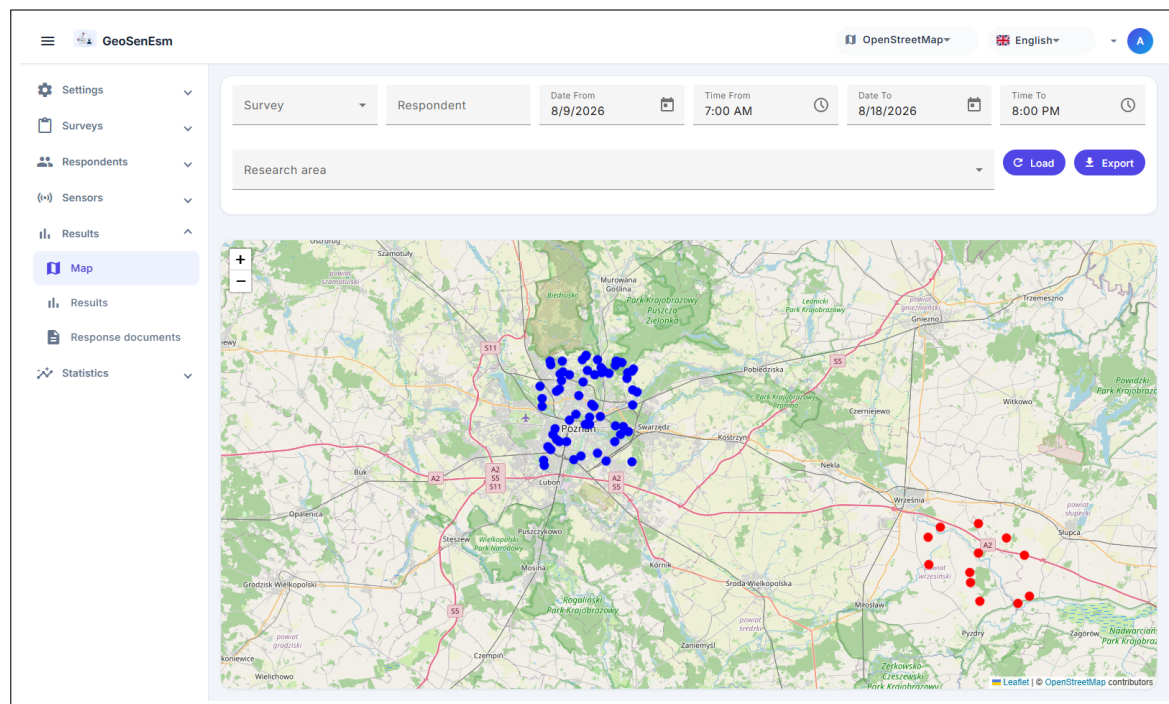


Figure 5.4: Map tab — GPS points shaded by inside vs. outside the research area.

5.5 Statistics

The Statistics section aggregates the operational data across the whole study and is the fastest way to get a feel for how the study is going. It is split into three view-toggles at the top of the page (*Global*, *Daily*, *Per participant*) and a dedicated *Daily completion* entry in the side menu. Every view shares a common KPI strip and a set of time-series charts underneath.

What “active” means here. Several KPIs and filters below — the Daily view’s *Active* KPI, Daily completion’s *Only respondents with set dates* filter, and Issues — all use the same definition of *active*: a respondent counts as active on a given day if they have a *survey window* assigned (Section 4.1.1) and that day falls within it (inclusive of the start and end dates). This is a fixed, per-respondent date range set by the administrator, not a rolling window measured backwards from today or from a respondent’s last submission — a respondent with no survey window assigned is never counted as active, regardless of how recently they submitted.

Whenever data is not yet available from the backend, a spinner is shown in place of the KPIs and charts. Switching views clears the stale values so the spinner is visible during every refresh.

5.5.1 Global view

Aggregates the whole study window into per-day time series (filled surveys, GPS points, sensor readings, participations outside the research area). It is the right place to look for weekly trends or unusual dips in data collection.

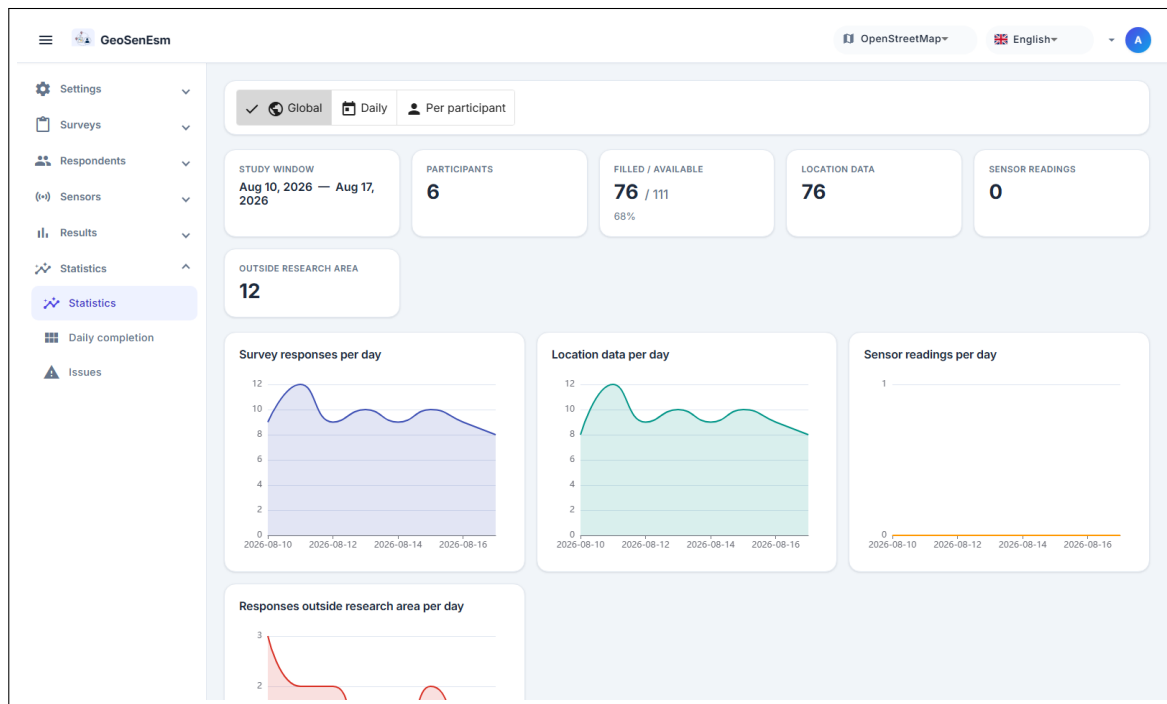


Figure 5.5: *Statistics — Global view.*

5.5.2 Per participant view

Focuses on a single respondent. The participant is picked from a **type-to-filter** input — type a substring of a username and the dropdown narrows to the matching accounts. Once a participant is chosen, the KPI strip and per-day time series update accordingly (Figure 5.6).

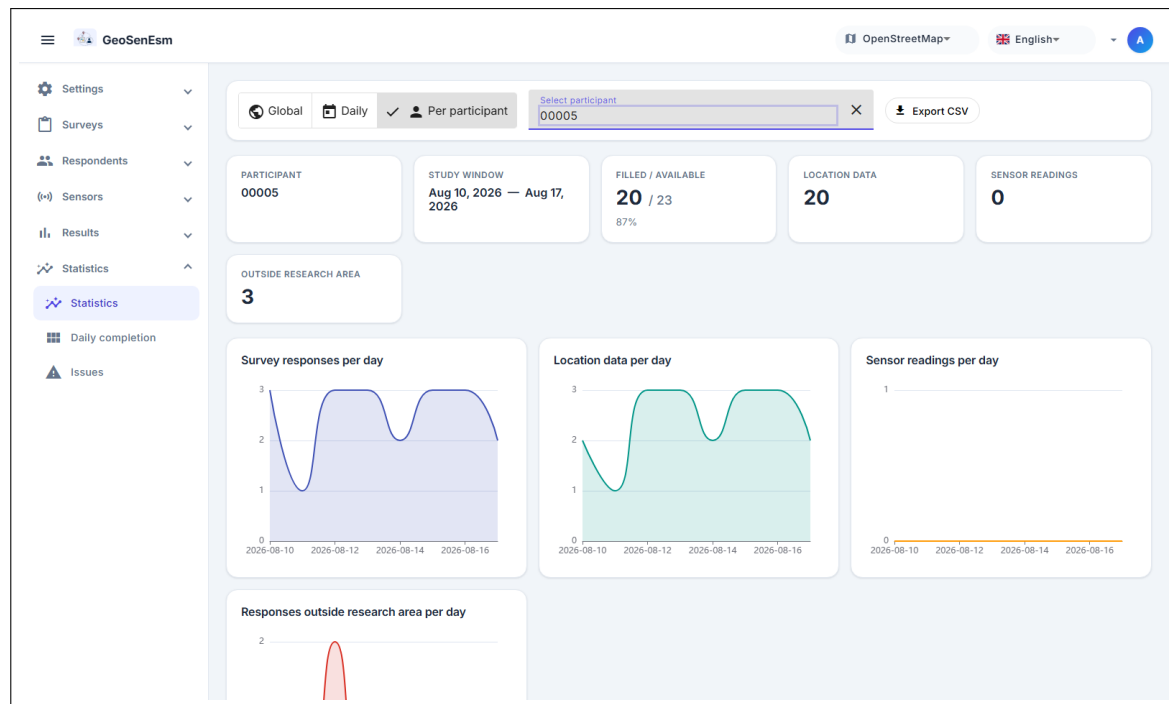


Figure 5.6: *Statistics — Per participant view with username autocomplete.*

5.5.3 Daily view

Zooms into a single calendar day, interpreted in each respondent's own local time. Instead of per-day series, the charts show per-hour buckets, which is useful when a slot schedule places several surveys inside the same day (Figure 5.7). The daily view also exposes an extra KPI, *Filled / Available active*, which narrows *Filled / Available* down to only the respondents who are active on the selected day (survey window covering that day — see above). Hovering the label opens a tooltip that explains the definition.

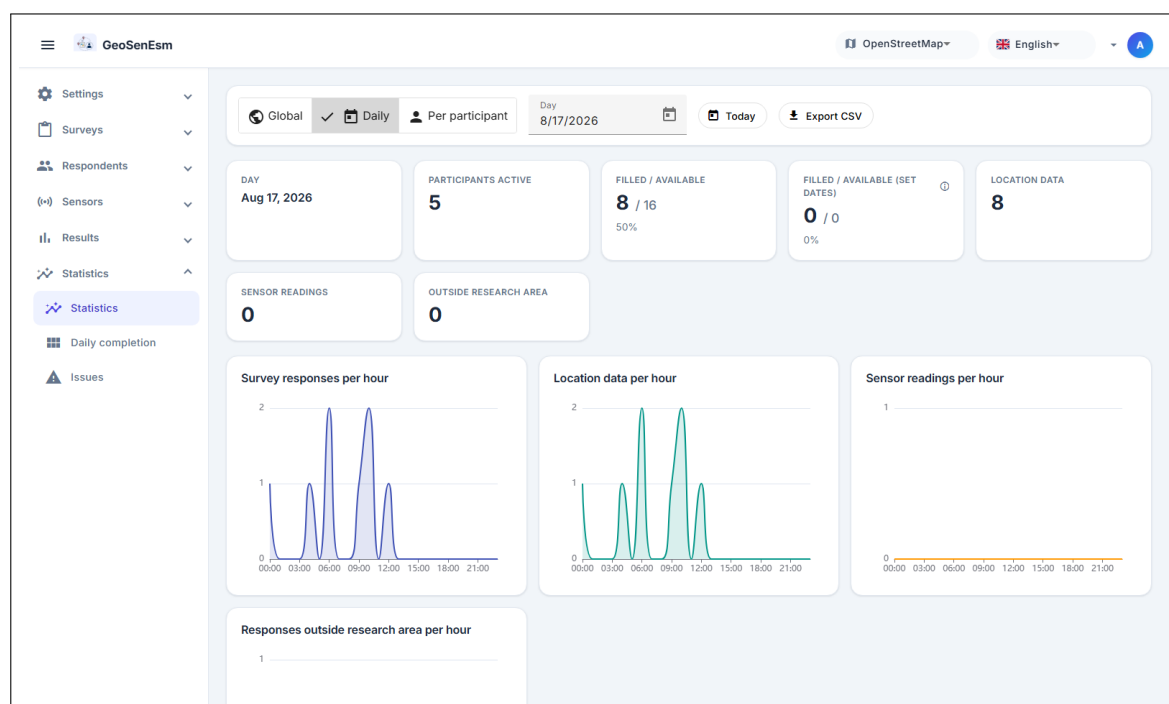


Figure 5.7: *Statistics — Daily view with hourly buckets and the “Active” KPI.*

5.5.4 Daily completion

Daily completion (Figure 5.8) is the compliance view. For a chosen day it draws one small card per respondent with a coloured stripe for every scheduled slot. The colour encodes what has been collected against that slot:

Daily-completion colour legend

- Submission collected with GPS *and* sensor data.
- Submission collected with GPS only.
- Submission collected with sensor only.
- Submission collected but neither GPS nor sensor data was linked.
- Slot has finished and no submission arrived.
- Slot has not started yet.

Each participant card can be resized: the toolbar exposes two sliders (width 110–220 px, height 90–160 px, step 5 px) with the selected size persisted in `localStorage` so the layout is stable across sessions.

The toolbar additionally offers an *Only respondents with set dates* filter, see Figure 5.9. When checked, the grid hides every respondent without a survey window, and further hides any respondent whose window does not cover the day chosen in the calendar — the same active-on-a-day rule used everywhere else in this section, not a rolling number of days.

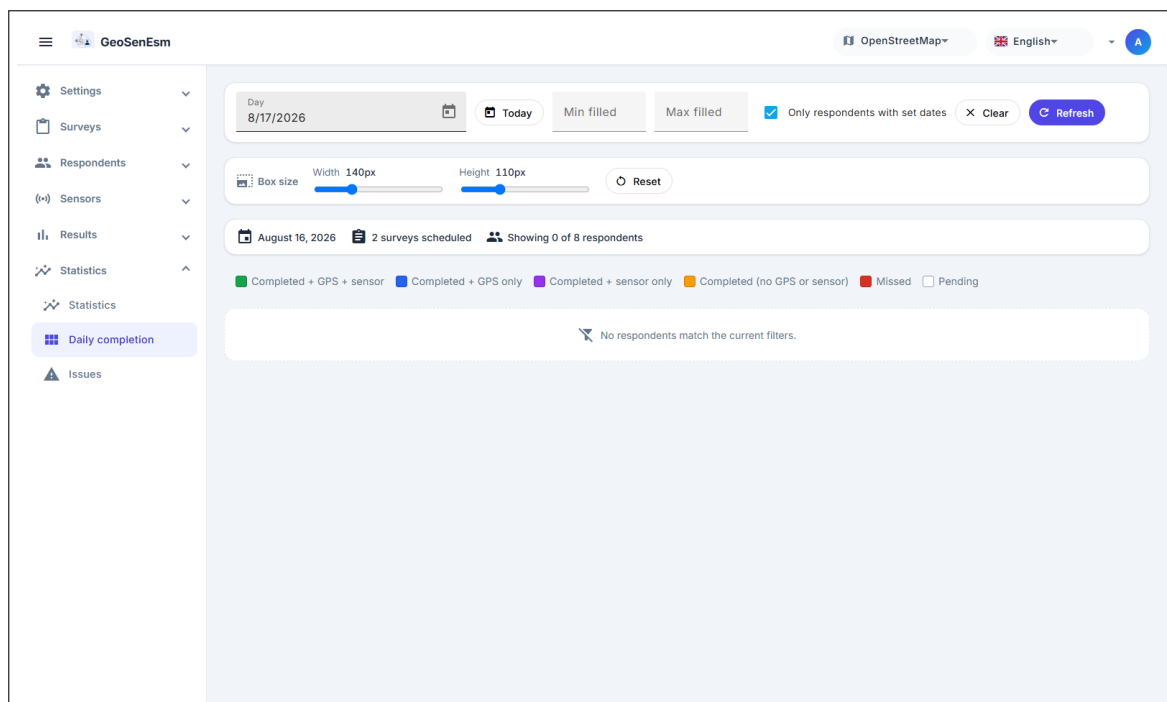


Figure 5.8: *Statistics — Daily completion.*

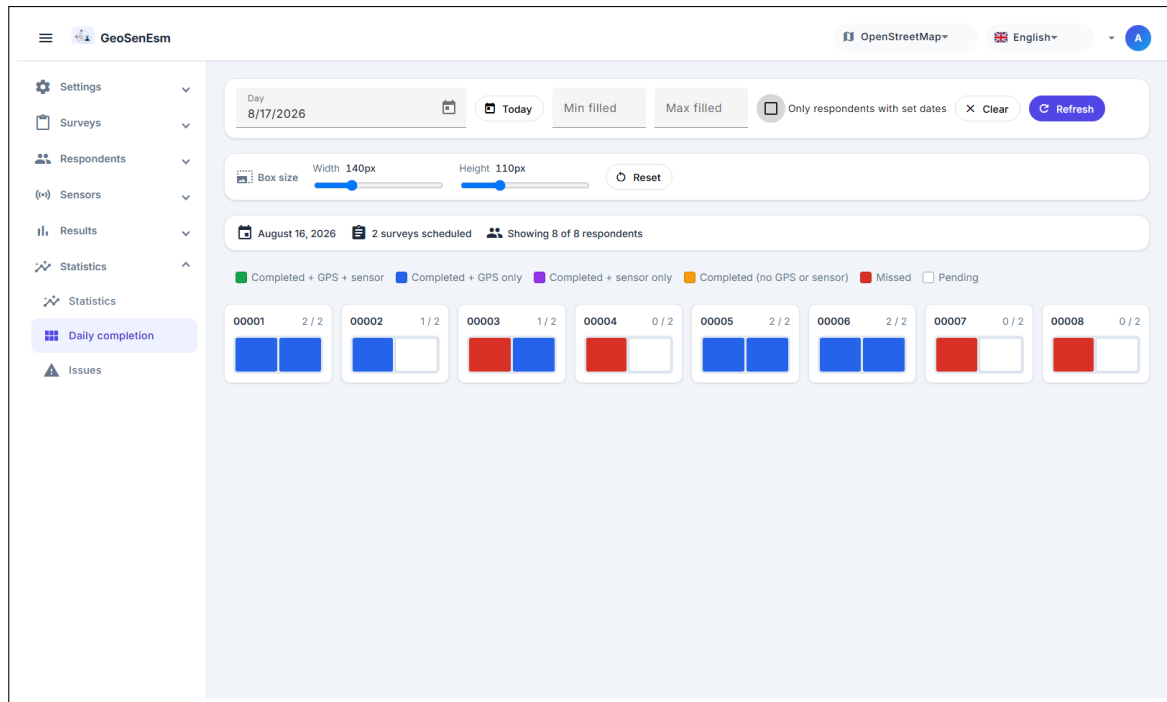


Figure 5.9: Daily completion with the “Only respondents with set dates” filter engaged — respondents with no survey window (or one that doesn’t cover the selected day) disappear from the grid.

5.5.5 Surveys outside research area

When a research-area polygon is defined, every GPS point that arrives is compared against it. Statistics includes a dedicated KPI card and time series that count submissions whose linked GPS row falls outside the polygon (Figure 5.10). Present on the Global, Daily and Per participant views, this number is a quick sanity check that respondents actually stayed within the study area.

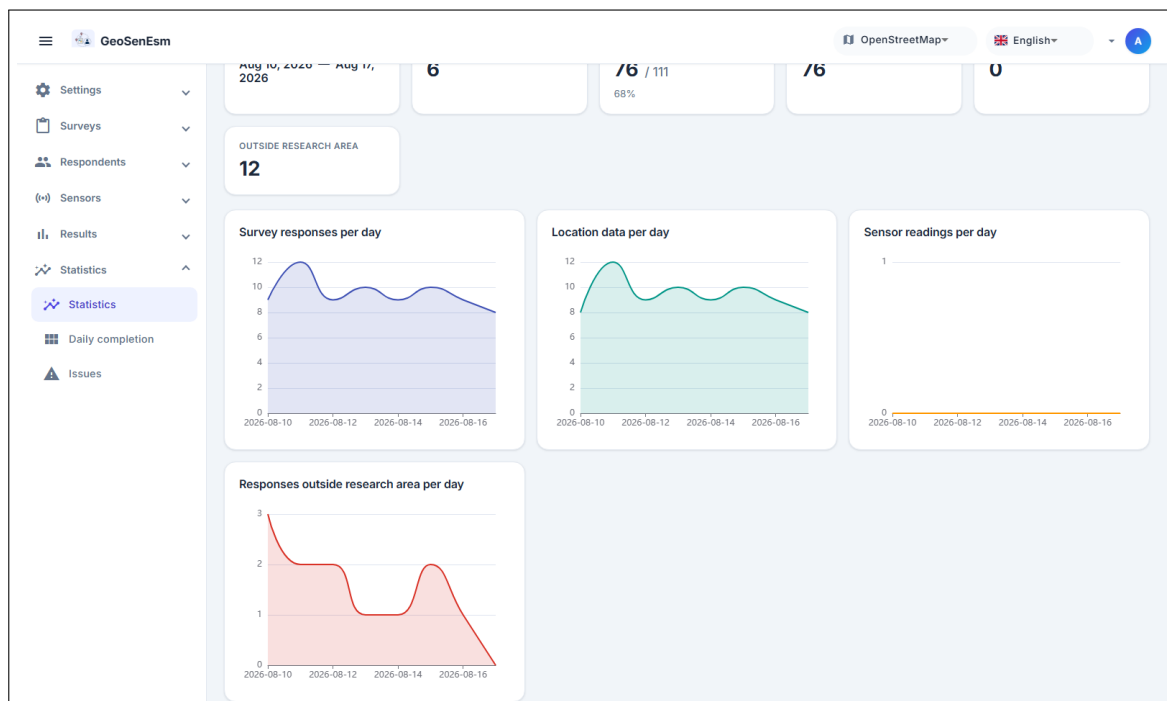


Figure 5.10: “Surveys filled outside research area” KPI and chart.

5.6 Issues

The Issues page (Figure 5.11) is new in v3.0.0 and answers a single question per respondent: *are they actually completing the study?* For each respondent with a survey window assigned (Section 4.1.1) — or for a custom date range — it computes the share of scheduled slots that were filled, the share with a linked GPS point, and the share with a linked sensor reading. Numeric filters (minimum skipped count, maximum survey/GPS/sensor fulfilment percentage) narrow the sortable, paginated table down to the respondents who most likely need a follow-up message or a hardware check.

The screenshot shows the 'Issues' page in the GeoSenEsm application. The left sidebar contains navigation links: Settings, Surveys, Respondents, Sensors, Results, Statistics, and Daily completion. The 'Issues' link is highlighted. The main content area has a title 'Issues' and a subtitle 'Find respondents with low survey, GPS, or sensor fulfillment inside their study window or a custom date range.' Below this is a yellow warning box stating: 'Custom dates only include respondents with an assigned survey window. Metrics use the overlap of your selected dates and each respondent's window; respondents with no surveys in that overlap are hidden.' There are two tabs: 'Survey window' (selected) and 'Custom dates'. A 'Load' button is next to the 'Custom dates' tab. Below the tabs are four filter boxes: 'Skipped at least', 'Survey fulfillment ≤ %', 'GPS fulfillment ≤ %', and 'Sensor fulfillment ≤ %'. There are 'Apply filters' and 'Clear filters' buttons. Below the filters, it says 'Showing 2 of 2 respondents'. A table displays the data for two respondents:

USERNAME	WINDOW	SURVEYS FILLED	SURVEY %	GPS %	SENSOR %	SKIPPED
00001	2026-08-09 → 2026-08-18	21 / 22	95.5%	90.9%	4.5%	1
00002	2026-08-09 → 2026-08-20	9 / 22	40.9%	40.9%	0.0%	13

At the bottom of the table, there is a pagination control showing 'Items per page: 25' and '1 - 2 of 2' with navigation arrows.

Figure 5.11: *Issues* — per-respondent survey, GPS and sensor fulfilment.

6 Mobile application

6.1 Login and first-run setup

The first screen the user sees is the login form (Figure 6.1). Respondents fill in the server URL (*API URL*, provided by the study administrator) together with their *username* and *password*. Accounts are created centrally in the admin panel and distributed to respondents out of band (email, print-out, and so on); self-registration is not possible. On login, the app also detects the device's time zone and reports it to the server, so every subsequent slot and notification time is computed in the respondent's actual local time (Section 4.2.2).

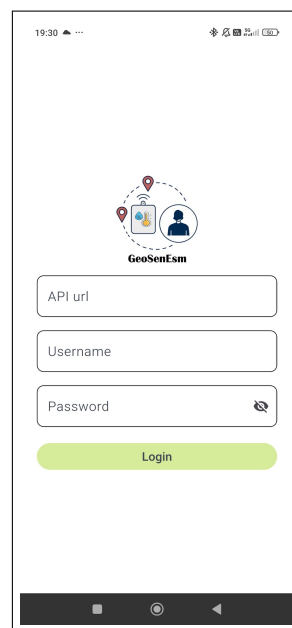
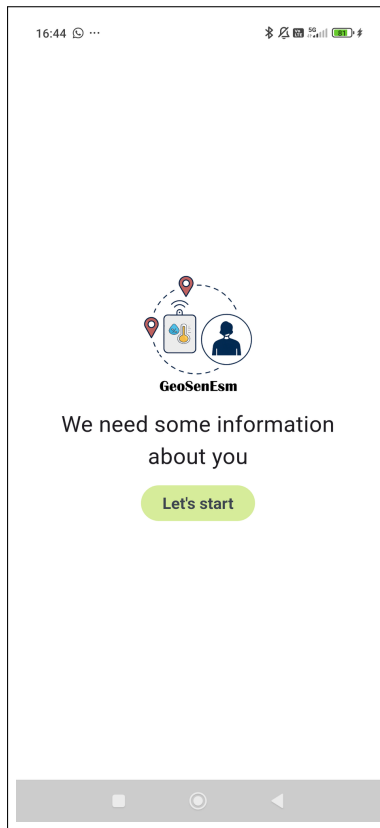
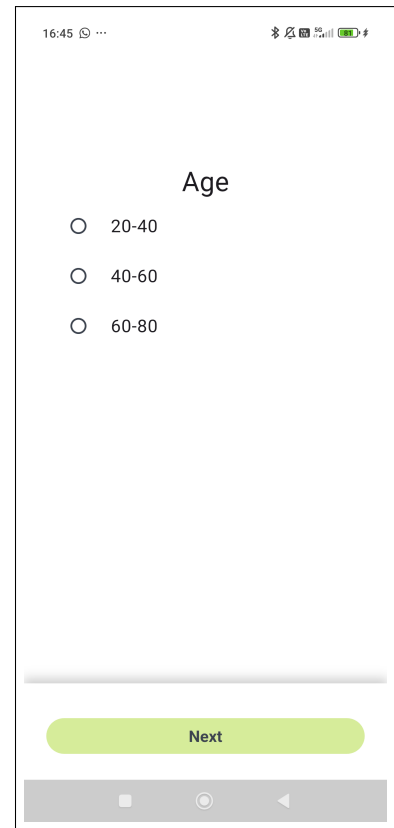


Figure 6.1: Login screen of the mobile app.

On the first launch the user is presented with the privacy policy. Sensor assignment is entirely server-driven: whenever the administrator has assigned a device configured through the sensor-profile engine (Section 3.3), the app syncs that assignment automatically in the background — there is no manual sensor picker or pairing step for the respondent to complete. Then the *initial survey* is shown (Figure 6.2); its answers place the respondent in a group that later surveys can target through section-level visibility rules.



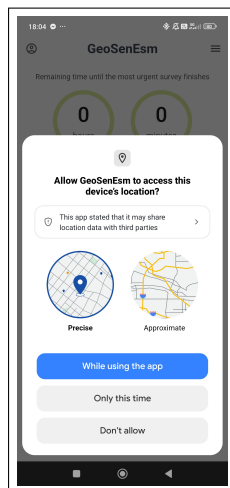
(a) Intro screen.



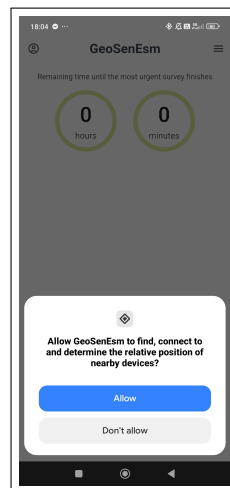
(b) Age question example.

Figure 6.2: Initial-survey example (language and age group).

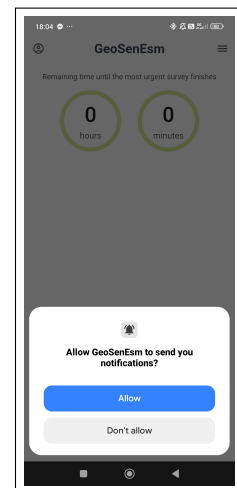
The app then requests the runtime permissions it needs. All three requests happen on first launch and are shown side by side in Figure 6.3.



(a) Location.



(b) Bluetooth.



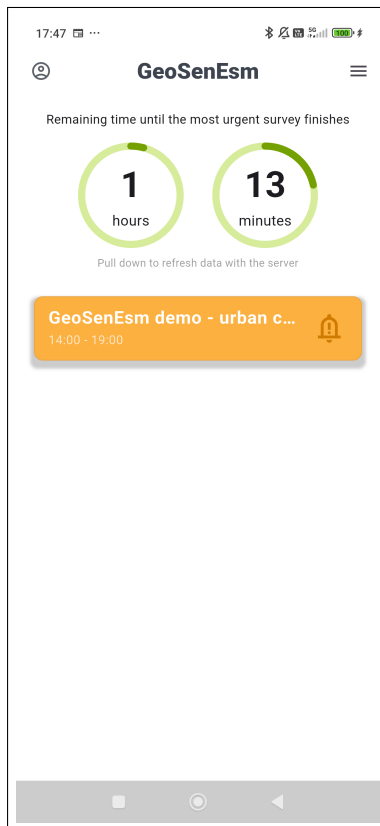
(c) Notifications.

Figure 6.3: First-run permission requests.

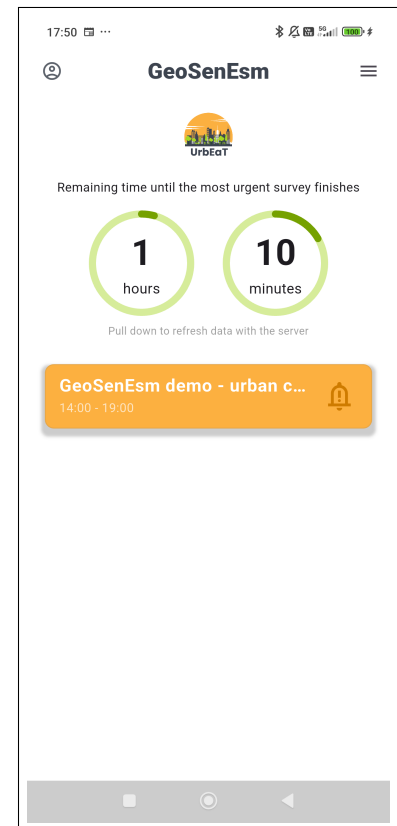
6.2 Home screen

The home screen (Figure 6.4) is intentionally minimalist. Bright orange buttons show the surveys that are currently open, together with a countdown to the slot's finish time. Navigation to

the menu and to the user profile lives in the top-left and top-right corners. A hint below the countdown reminds the respondent that pulling down refreshes the survey list from the server. When the administrator has uploaded a study logo (Section 2.1), it replaces the default text wordmark at the top of the screen. The layout respects the device's font-size accessibility setting.



(a) Default wordmark.

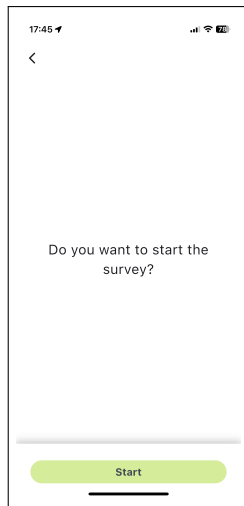


(b) Custom study logo.

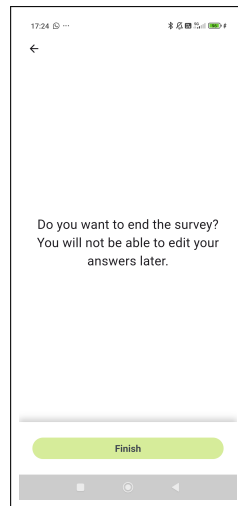
Figure 6.4: Home screen of the mobile app.

6.3 Filling in a survey

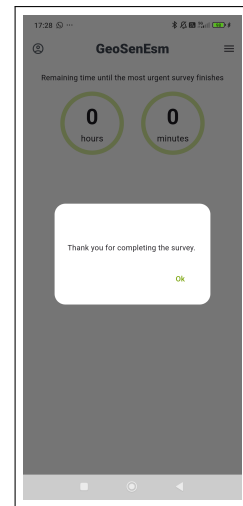
1. **Open the app** and tap the orange button whose caption matches the active survey. The countdown next to the button shows how long the slot is still open. Bluetooth should be enabled: the app tries to read a fresh reading from the assigned sensor while the survey is being filled in.
2. **Introduction, end and submission screens** (Figure 6.5) confirm the start of the survey, ask for confirmation before ending it, and confirm that responses were submitted. Once submitted, a response cannot be edited.
3. If a used sensor-data parameter is not covered by any connected sensor's reading, the app prompts the respondent to enter a value manually before the response is submitted (Figure 6.6) — every used parameter is guaranteed this fallback (Section 3.4).



(a) Start prompt.



(b) End prompt.



(c) Submission confirmation.

Figure 6.5: Confirmation screens shown before, during and after a survey.

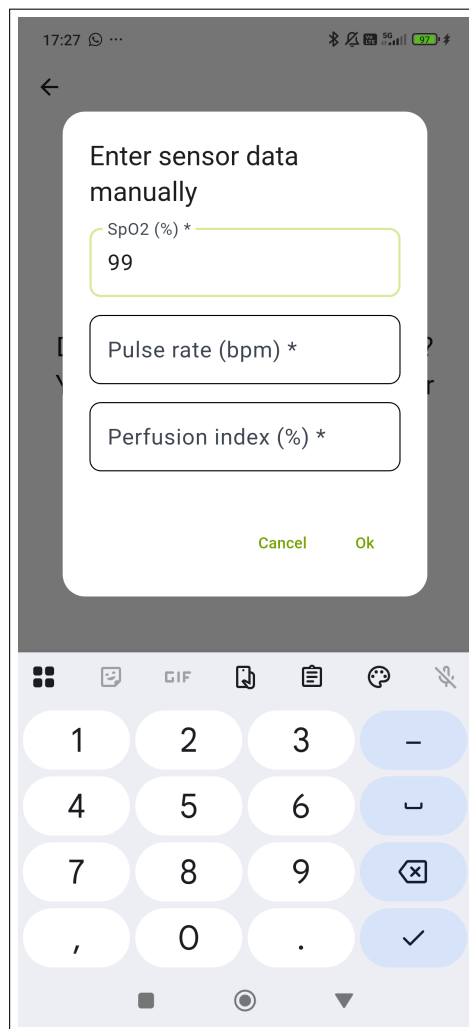


Figure 6.6: Manual entry prompt for a sensor-data parameter no connected sensor covered (pulse oximeter example).

6.4 Question types

Surveys support seven question types (Figure 6.7).

(a) Single choice.

(b) Multiple choice.

(c) Linear scale.

(d) Yes / No.

(e) Number input.

(f) Text input.

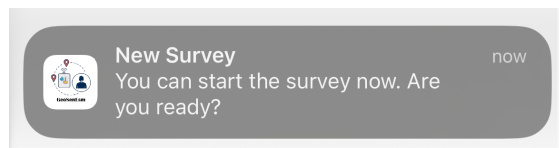
(g) Image choice.

Figure 6.7: The seven question types supported by GeoSenEsm.

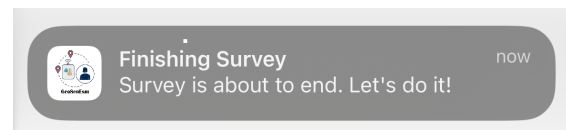
On submission the app sends the response to the server together with the matching location and sensor reading.

6.5 Notifications

Respondents are notified according to the notification rules the administrator configured for that survey (Section 4.2.2); by default, that means once when a slot opens and again 15 minutes before it closes (Figure 6.8), computed in the respondent's own local time.



(a) Slot opens.



(b) Slot closes soon.

Figure 6.8: Push notifications sent by the mobile app.

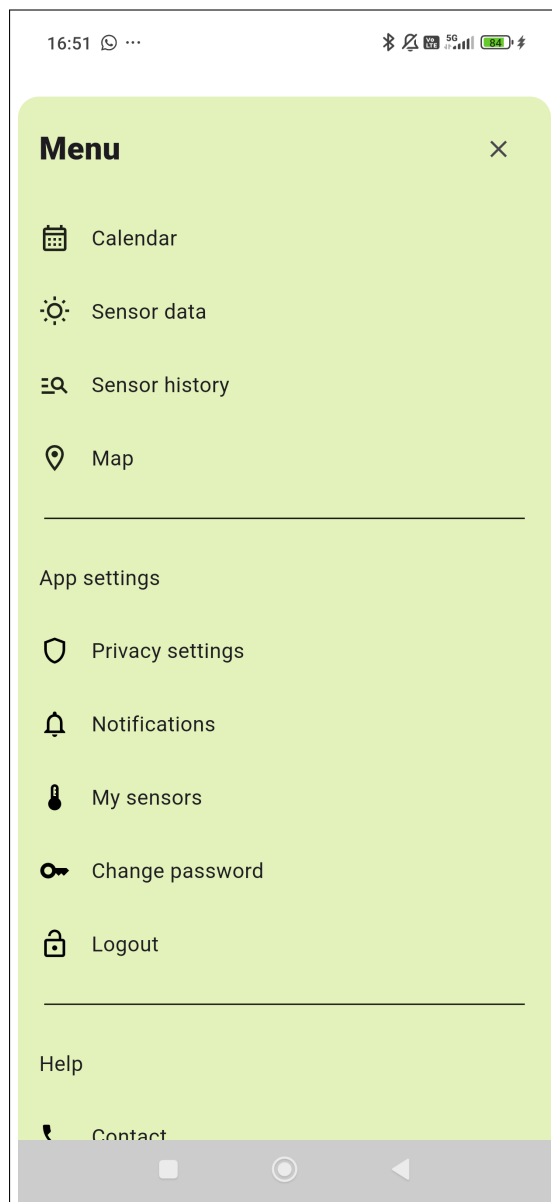
6.6 Offline mode

The app works both online and offline. Once a device has synced with the server, all published surveys are cached locally. Respondents can complete them without a network connection;

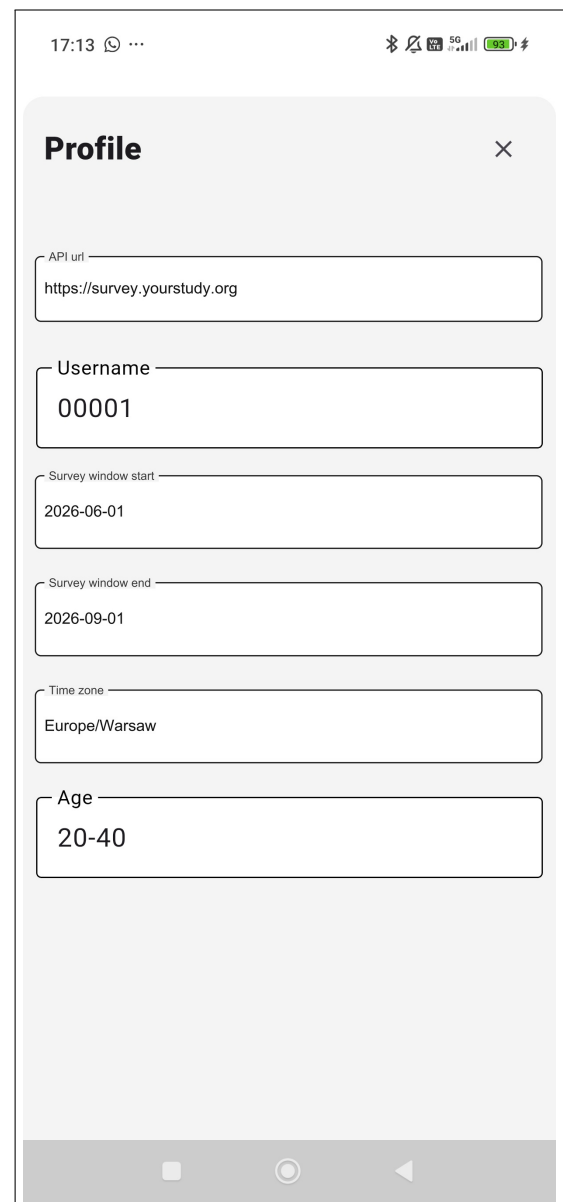
answers, GPS points and sensor readings are pushed to the server as soon as connectivity returns.

6.7 Menu and app settings

The hamburger menu in the top-right corner of the home screen (Figure 6.9a) is the entry point to every module described below (Calendar, Map), the sensor modules when the study has them enabled, and a second group of account-level settings described below. Tapping the account icon in the top-left corner instead opens the respondent's own *Profile* (Figure 6.9b): a read-only summary of the API URL, username, survey window and time zone recorded for the account, together with their initial-survey answers.



(a) Menu.

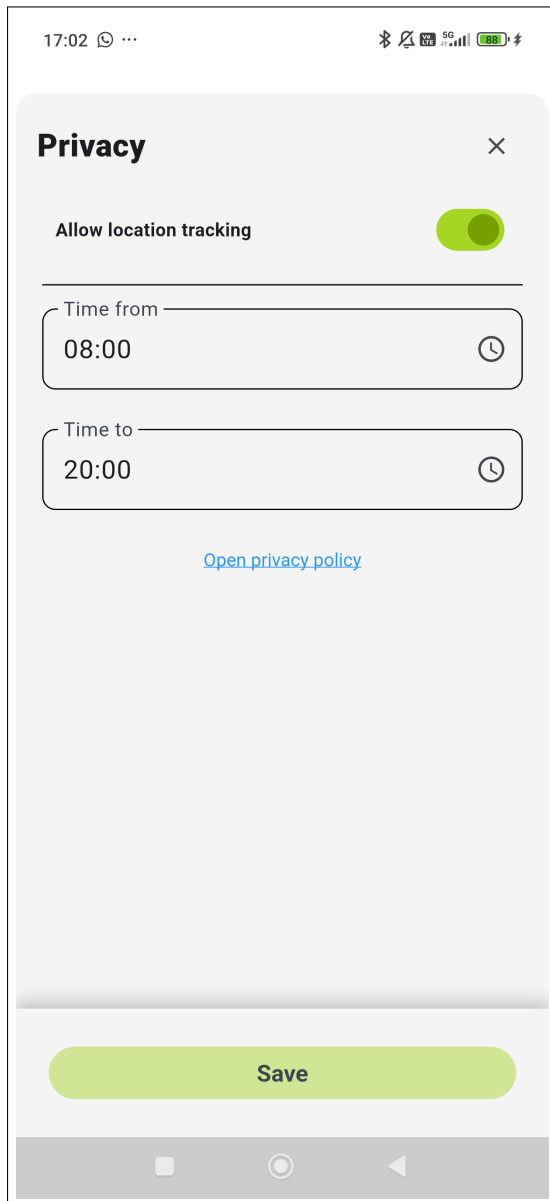


(b) Profile.

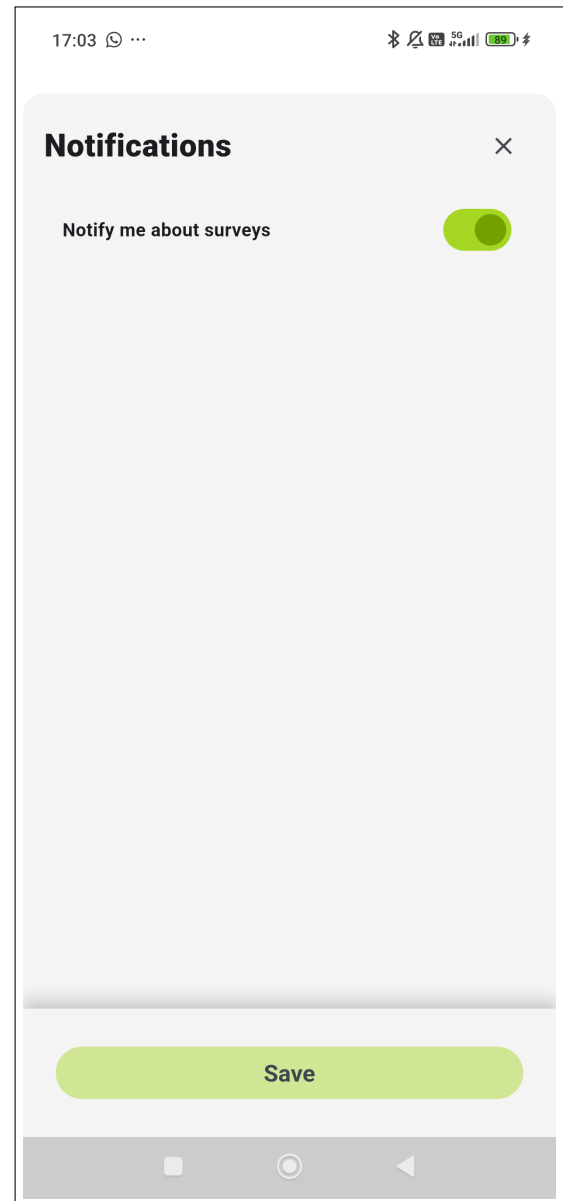
Figure 6.9: Menu and profile screens.

Under *App settings* the respondent can:

- Restrict **location tracking** to a daily time window, or turn it off entirely (Figure 6.10, left).
- Turn survey **notifications** on or off (Figure 6.10, right).
- Review **My sensors** (Figure 6.11): every sensor type currently assigned to the account and the parameters each one reads.
- **Change password** (Figure 6.12, left) or **log out**. Logging out warns that any response not yet uploaded to the server will be lost (Figure 6.13).
- Reach the study's **contact** organizer directly from the app (Figure 6.12, right).



(a) Privacy.



(b) Notifications.

Figure 6.10: Privacy and notification settings.

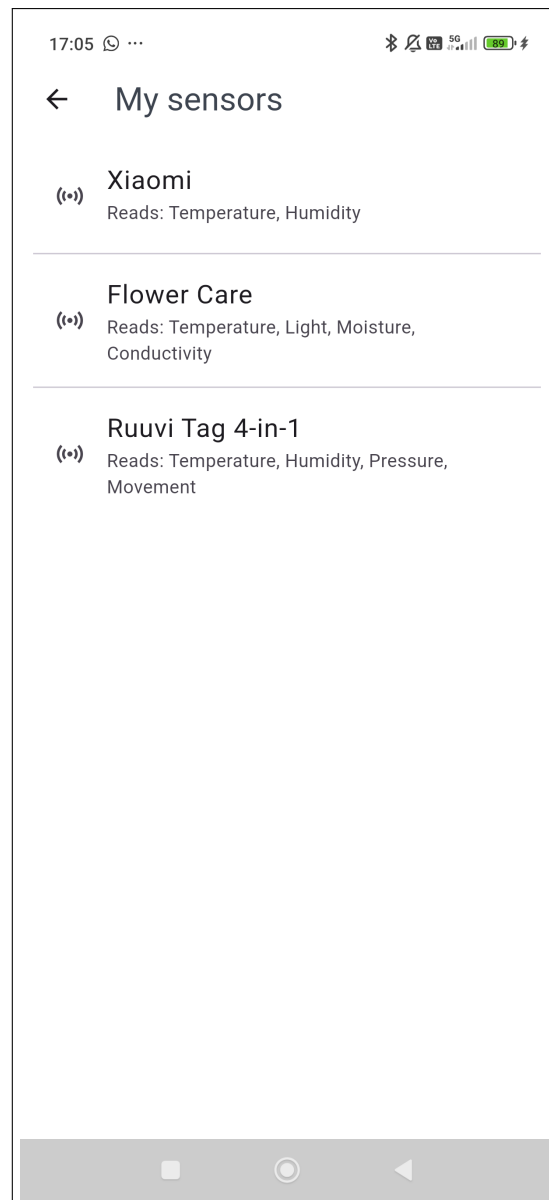
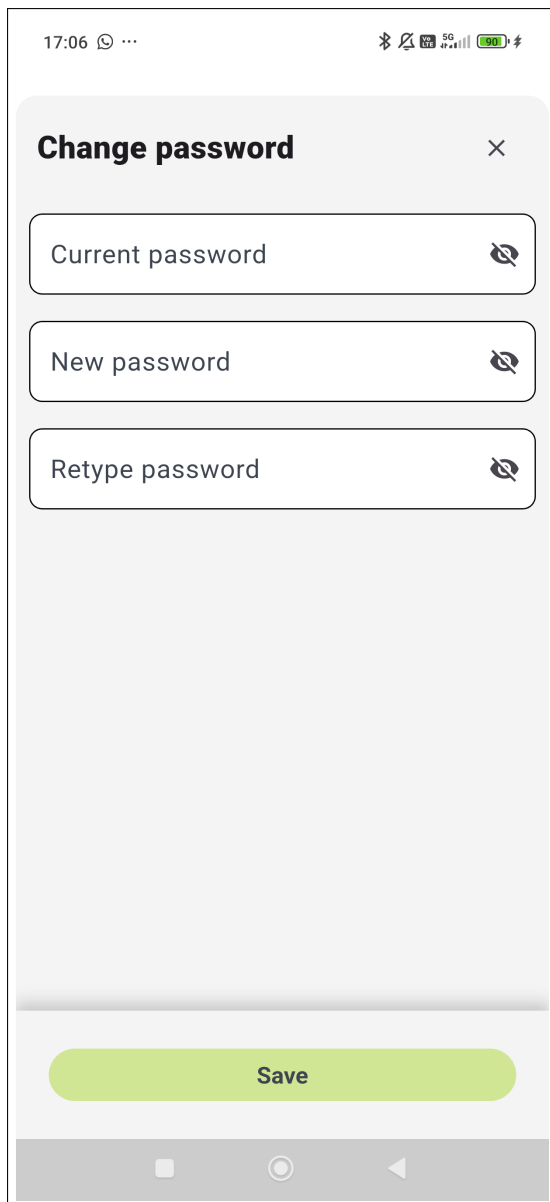


Figure 6.11: *My sensors* — devices currently assigned to this account.



17:06 ⓘ ...

Change password ×

Current password 

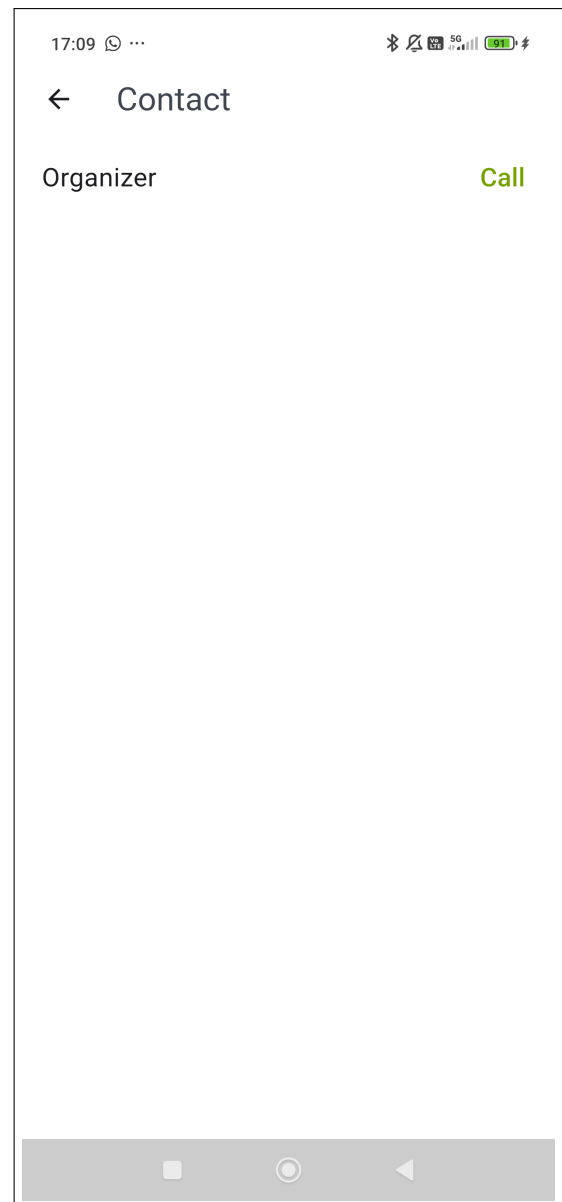
New password 

Retype password 

Save

This screen is for changing a password. It features a title bar with a close button (×). Below the title are three input fields: 'Current password', 'New password', and 'Retype password'. Each input field has an eye icon to toggle visibility. At the bottom, there is a green 'Save' button. The status bar at the top shows the time as 17:06 and various system icons.

(a) Change password.



17:09 ⓘ ...

← **Contact**

Organizer **Call**

This screen displays contact information. It has a title bar with a back arrow and the title 'Contact'. Below the title, the name 'Organizer' is shown, followed by a green 'Call' button. The status bar at the top shows the time as 17:09 and various system icons.

(b) Contact.

Figure 6.12: Change password and contact screens.

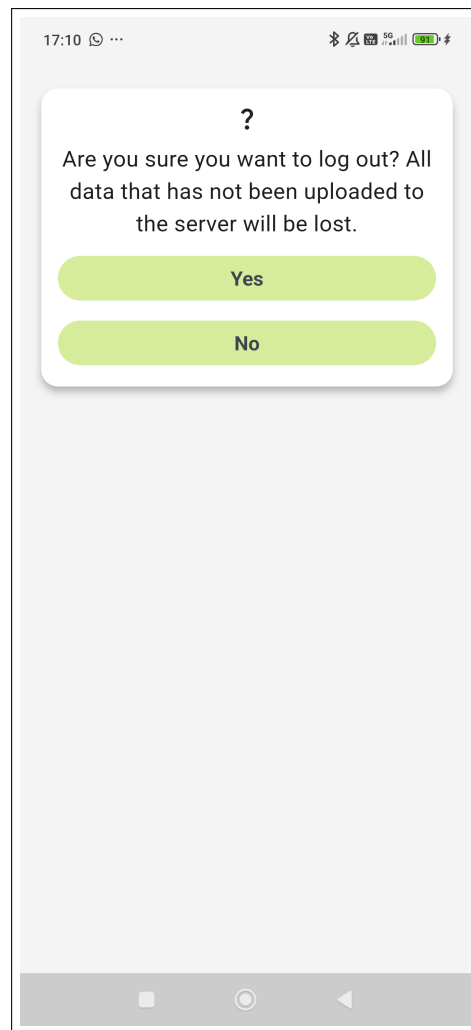


Figure 6.13: Logout confirmation — unsynced data would be lost.

6.8 Additional views

GeoSenEsm exposes what the app has captured to the respondents themselves, building trust and giving them a way to spot problems (for example, a Bluetooth issue that stopped sensor readings for a week). These are reached from the menu described above.

6.8.1 Map

The map view (Figure 6.14, left) plots every submitted location point. Tapping a marker opens its details (right).

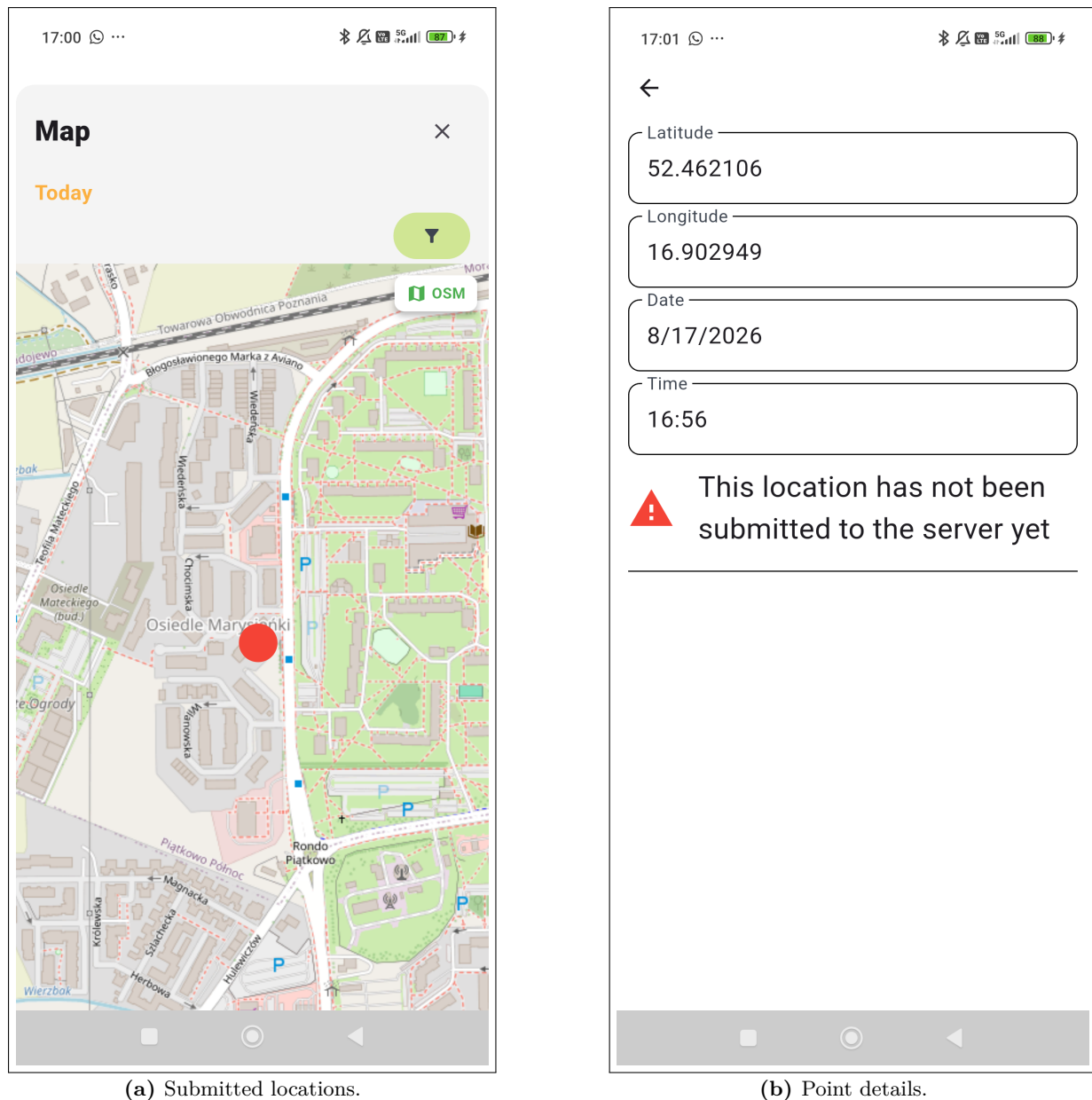
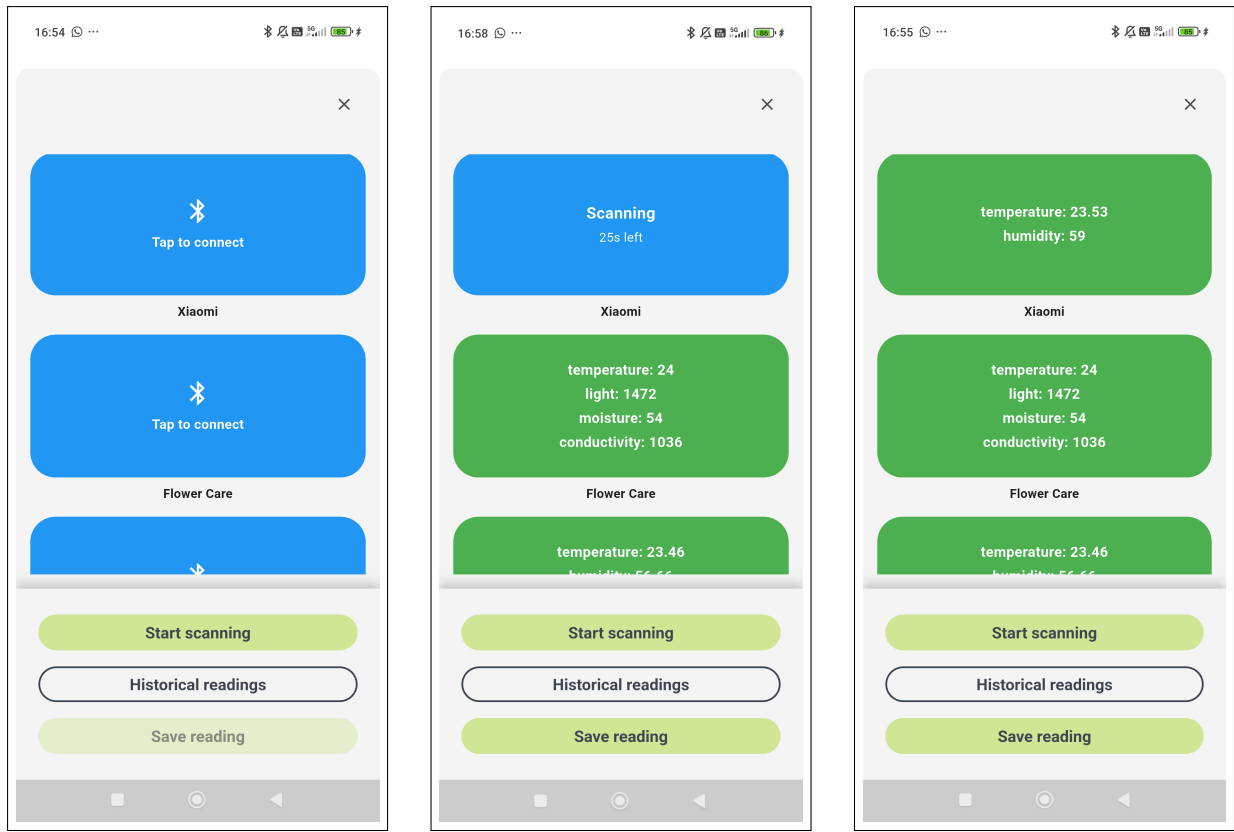


Figure 6.14: Mobile map module.

6.8.2 Sensor data and history

The sensor module shows one card per assigned sensor type, so a respondent with more than one sensor (for example a temperature sensor and a pulse oximeter) connects to and reads all of them from the same screen instead of one at a time (Figure 6.15); the history sub-view lists every reading the app has captured (Figure 6.16). Both views are driven by whichever parameters the assigned sensor profile(s) report — typically temperature and humidity, but SpO₂ and pulse rate for a pulse oximeter, or moisture and light for a plant sensor, without any change to the app itself.



(a) Before scanning.

(b) Scanning.

(c) Live readings.

Figure 6.15: Sensor data screen, one card per assigned sensor type.

Date	Source	Values
8/17/2026 16:56	ruuvi	temperature: 56.66, pressure: 1013.25, movement: 40
8/17/2026 16:56	xiaomi	temperature: 59.00
8/17/2026 16:56	flower_care	temperature: 54.1, moisture: 1036

Figure 6.16: Reading history, across every assigned sensor.

6.8.3 Progress calendar

The calendar (Figure 6.17) colour-codes the respondent's participation history:

- ■ Green — surveys completed on time.
- ■ Red — surveys missed.
- ■ Yellow — surveys still upcoming.

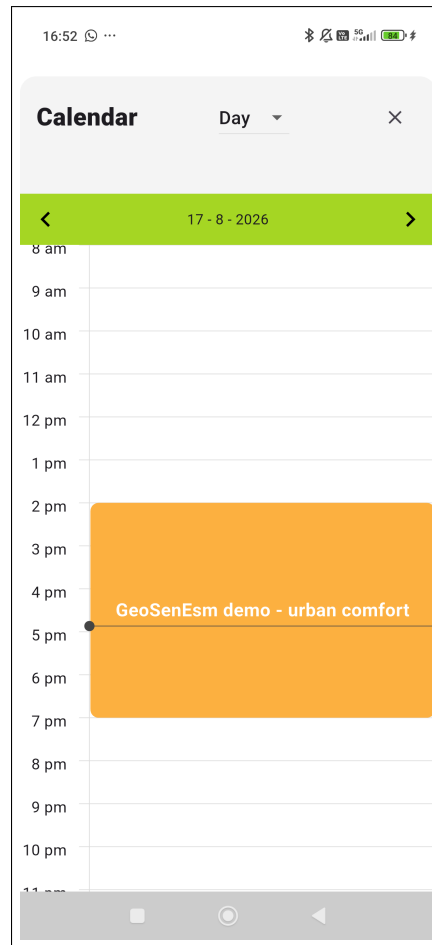


Figure 6.17: Colour-coded participation calendar.

6.9 Design consistency

Surveys deliberately avoid complex gestures such as swiping or drag-and-drop. One question is shown per page and can be answered without scrolling. The colour palette (orange, green, dark grey) and the study logo are used consistently across the mobile app and the admin panel.

7 Closing notes

The design of GeoSenEsm follows a few consistent principles:

- *Prioritise the respondent.* Keep the mobile UI simple, the permissions transparent and the sensitive data controllable.
- *Meet the researcher where they are.* Every artefact (results, response documents, GPS, sensor readings) is exportable to CSV, JSON or ZIP so it can flow straight into existing analysis pipelines, in the researcher's own CSV convention.
- *Configuration over code.* Adding a new Bluetooth sensor, tuning a notification schedule, or changing what counts as "sensor data" are all admin-panel actions, not mobile-app releases.
- *Make problems easy to spot.* The Statistics dashboard, the Daily completion grid, the Issues page and the outside-research-area KPI are all there so an issue with a single respondent or a single day of the study is visible at a glance.

The Experience Sampling Method integrates naturally with a mobile application: participants carry their phone with them at all times, and the app can capture context (location, environmental or biometric sensor readings) alongside their subjective response. Cross-referencing the two lets researchers build a richer picture of how urban environments and physiological state shape everyday behaviour.